

Inverse Problem Approach for Non-Perturbative QCD: Theoretical Foundation

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Abstract

We propose a novel theoretical framework, the inverse problem approach, to calculate the non-perturbative QCD quantities. Using the dispersion relation of quantum field theory, it solves the unknown low-energy non-perturbative quantities by the inverse problem, with the known high-energy perturbative calculations as inputs. We prove that the inverse problem of dispersion relation is ill-posed, with the solutions being unique but unstable. Based on the state-of-the-art mathematics of inverse problems, the Tikhonov regularization method is used to obtain stable approximate solutions which converge to the true values of solutions as the input uncertainties approach zero. We have tested some toy models to vividly show the main features of the inverse problem approach. It can be found that this approach does not have any artificial assumptions and can systematically improve the precision of the solutions.

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1 Introduction

The strong interaction, one of the four fundamental forces in nature, governs the majority of subatomic phenomena and is described by Quantum Chromodynamics (QCD). At high-energy scales, the QCD coupling constant is sufficiently small to allow physical quantities to be calculated perturbatively, resulting in highly successful theoretical predictions. However, at low-energy scales, the coupling constant grows significantly, rendering perturbative methods inapplicable. This presents one of the most formidable challenges in theoretical physics, known as the non-perturbative problem of strong interactions.

The non-perturbative challenges of QCD is one of the most fundamental barriers to advancing particle physics and nuclear physics. It remains a central unsolved puzzle to understand the dynamical mechanism of color confinement and hadronization, which governs how quarks bind into observable hadrons such as protons and neutrons. The spatial or momentum distributions of quarks and gluons within hadrons, which are crucial for interpreting high-energy scatterings, are poorly understood such as the generalized parton distribution functions (GPDs), the transverse-momentum dependent parton distribution functions (TMDs), the light-cone distribution amplitudes (LCDAs), and the fragmentation functions (FFs). The internal structures of exotic hadronic states observed in experiments are debated by compact multiquarks, molecular states, hybrid states, or nonresonant kinematic effects. The properties of QCD phase transitions and the precise location of the critical point in the nuclear matter phase diagram remain significant challenges in theories and experiments. Moreover, indirect probes of new physics beyond the Standard

Model are intrinsically tied to non-perturbative QCD effects. For instance, the dominant theoretical uncertainties in the muon anomalous magnetic moment and the rare decays of $B \rightarrow K^{(*)} \ell^+ \ell^-$ arise from the hadronic vacuum polarization and the hadronic transition form factors, respectively.

Over the past half century, several non-perturbative methods have been well developed and extensively utilized. Each of them has its advantages and disadvantages. Lattice QCD [1], based on the discretized Euclidean spacetime, is regarded as the most reliable and rigorous first-principles approach for non-perturbative calculations, whereas it requires substantial computational resources and faces challenges in addressing real-time dynamics or properties of excited states. QCD sum rules [2, 3] and the Dyson-Schwinger equations [4] are broadly employed non-perturbative methods in the continuum Minkowski spacetime, but suffer some inevitable systematic uncertainties from the quark-hadron duality and truncation scheme, respectively. In addition, phenomenological models, including various quark models [5], are inherently model-dependent, significantly constraining their predictive power. As a result, the development of novel non-perturbative computational methods is always highly valued and welcomed in the field.

In recent years, a novel non-perturbative QCD method was developed, the inverse problem approach. It was first proposed in studies of the $D^0 - \bar{D}^0$ mixing in 2020 [6], where the charm-quark expansion does not converge well, leading to perturbative calculations of the mass and width differences of neutral charmed mesons are several orders of magnitude smaller than the experimental measurements. Taking the charm-quark scale as non-perturbative, the relevant quantities can be obtained by solving the inverse problem of an integral equation from dispersion relation, with the input of high-energy perturbative calculations by the effective theory of heavy-quark expansion above the bottom-quark scale. The theoretical results of the mixing parameters of $D^0 - \bar{D}^0$ are then consistent with the experimental data [6]. Subsequently, this approach has been extended to investigate a variety of physical problems [7–11], with expansion of polynomials and inverse matrix method. Firstly, charm physics is usually difficult in theory, since the charm-quark scale is neither high enough for the heavy-quark expansion nor low enough for the chiral perturbation theory. It can now benefit from the inverse problem approach as seen in $D^0 - \bar{D}^0$ mixing. Secondly, the whole non-perturbative region is solved simultaneously in this approach, so that the properties of excited states can be determined together with the ground state. The study of excited states is always a challenge for most QCD-inspired approaches, yet it represents a significant advantage of the inverse problem approach. Thirdly, the predictive power of QCD sum rules is significantly limited by large uncertainties, primarily stemming from the assumption of global quark-hadron duality. This assumption neglects the excited states and continuum spectrum, thereby introducing a free parameter that cannot be determined from the first principles of QCD. The inverse problem approach offers a significant

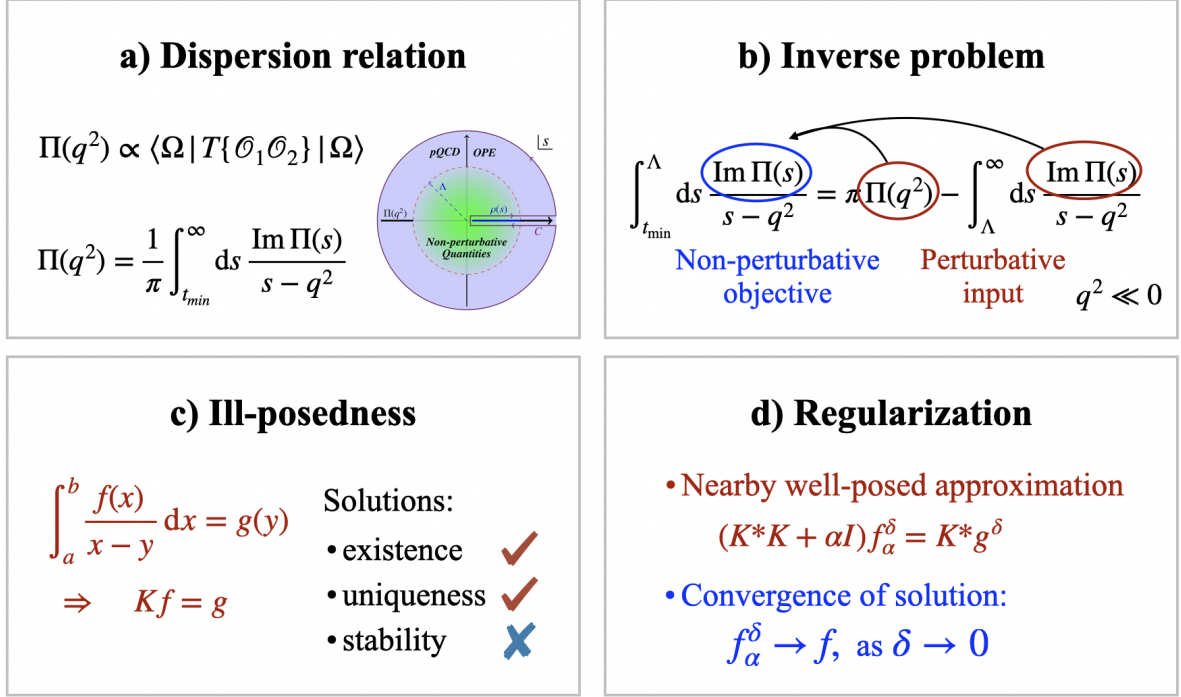


Figure 1: Illustration of the inverse problem approach of dispersion relation.

improvement to QCD sum rules by directly solving the excited states and continuum spectrum, thus avoiding the model-dependent free parameter. This can increase the predictive capability and reliability of the method.

However, the mathematical structure of the inverse problem approach in previous works has remained unclear, which is one of the main purposes of this work. Here, we systematically develop the theoretical framework of the inverse problem approach, aiming at building a solid mathematical foundation. The main ideas of the inverse problem approach are as follows, which is also illustrated in Fig. 1. It is based on an integral equation derived from the dispersion relation in quantum field theory, where the high- and low-energy scales are separated into distinct components. Then the unknown non-perturbative quantities, represented as the integrand, are determined by solving an inverse problem, with the known perturbative calculations as input. The inverse problem is a well-established field in mathematics, whose practical application generally requires a thorough investigation of its mathematical properties, particularly its inherent ill-posedness. We prove that the inverse problem of the dispersion-relation integral equation is ill-posed, with unique but unstable solutions. To overcome the ill-posed problem, regularization methods have been developed in mathematics to obtain stable approximate solutions which converge into the true values as the uncertainties of inputs diminish and approach zero.

In this work, we employ the state-of-the-art Tikhonov regularization method, which is well known for its straightforward mathematical properties and particular friendliness towards beginners. The

regularization parameter is determined using well-established and intuitive mathematical methods. The inverse problem approach is firmly grounded in rigorous mathematical principles, without any artificial assumption. Several toy models are examined to illustrate the key features of the inverse problem approach. The results demonstrate that solution precision can be systematically improved by this approach as input errors decrease, as well as by employing suitable regularization or combining it with high-precision experimental or lattice QCD data.

This paper is organized as follows. In Sec. 2, we give a brief introduction to the inverse problem of the dispersion relation. In Sec. 3, we prove the uniqueness and instability of the inverse problem. In Sec. 4, the regularization method are introduced and used to overcome the ill-posedness. In Sec. 5, some toy models are discussed to demonstrate the effectiveness of regularization method and illustrate the main features of the inverse problem approach. Finally, some physical discussions and conclusions are given in Sec. 6. For the convenience of the reader, key mathematical concepts are presented in Appendix A.

2 Inverse Problem of Dispersion Relation

The inverse problem approach for dispersion relations stems from the fact that they unify the perturbative and non-perturbative contributions within a single integral equation. In this section, we will introduce the inverse problem of dispersion-relation integral equations.

Dispersion relations have been widely used in particle physics and nuclear physics. The QCD sum rules approach is based on dispersion relations, which allow the calculation of hadronic properties using perturbative operator product expansion techniques [2]. Dispersion relations also play a key role in hadron-hadron scattering, particularly in chiral perturbation theory [12–14]. The nucleon charge radius can be determined via dispersion relations, which relates the full form factors to the integration of their imaginary parts [15]. Dispersion relations are also crucial in indirect searches for new physics beyond the Standard Model, such as analyzing charm-quark loop effects in $B \rightarrow K^* \ell^+ \ell^-$ decays [16] and determining hadronic vacuum polarization contributions to muon g-2 through a data-driven approach [17]. The core principle of dispersion relations is to determine non-perturbative hadronic quantities by leveraging perturbative calculations or experimental measurements.

Dispersion relations arise from the causality and hence the analytic properties of quantum field theory (QFT). We outline the derivation of a general dispersion relation, which starts from a well-defined quantity in QFT, such as a correlation function

$$\Pi(q^2) = i \int d^4x e^{iqx} \langle \Omega | T \{ O_1(x) O_2(0) \} | \Omega \rangle. \quad (2.1)$$

This two-point correlation function is simple and convenient for discussion, without the explicit expres-

sions of Lorentz structures. It involves the time-ordered product (T) of two local operators at distinct four-dimensional spacetime points, $O_1(x)$ and $O_2(0)$, evaluated in the QCD vacuum state $|\Omega\rangle$. $\Pi(q^2)$ is defined in the momentum space.

Then we consider all singularities of the correlation function in the complex plane. In QCD, the complex plane is free from singularities except along the positive real axis, as illustrated in a) of Fig. 1. Within the region enclosed by the contour, the correlation function $\Pi(s)$ is analytic. Applying the Cauchy integral formula of complex functions, we derive the following dispersion relation:

$$\begin{aligned}\Pi(q^2) &= \frac{1}{2\pi i} \oint_C ds \frac{\Pi(s)}{s - q^2} \\ &= \frac{1}{2\pi i} \oint_{|s|=R} ds \frac{\Pi(s)}{s - q^2} + \frac{1}{2\pi i} \int_{t_{\min}}^R ds \frac{\Pi(s + i\epsilon) - \Pi(s - i\epsilon)}{s - q^2},\end{aligned}\quad (2.2)$$

where R is the radius of the large circle in a) of Fig. 1, taken to be infinitely large, and $t_{\min}$ denotes the threshold of the lowest singularities. Here, $\epsilon > 0$ is an infinitesimal quantity introduced to shift the path slightly above or below the positive real axis.

In principle, $\Pi(s)$ may not vanish as $s \rightarrow \infty$ in many real physical problems. In such cases, the use of subtraction or derivative dispersion relations becomes necessary, which renders the first term on the right-hand side (R.H.S) of Eq. (2.2) being vanishing. In this work, we restrict our analysis to the simplest case where $\Pi(s) \rightarrow 0$ as $s \rightarrow \infty$, ensuring that this term vanishes. The proofs and discussions in this article are nonetheless applicable to other forms of dispersion relations. Applying the Schwarz reflection principle, which gives $\Pi(s + i\epsilon) - \Pi(s - i\epsilon) = 2i \text{Im } \Pi(s)$ for $s > t_{\min}$, the dispersion relation becomes

$$\Pi(q^2) = \frac{1}{\pi} \int_{t_{\min}}^{\infty} ds \frac{\text{Im } \Pi(s)}{s - q^2}. \quad (2.3)$$

With $q^2 < 0$, it is widely used in QCD or light-cone sum rules [2,3,18,19], as well as in relating space-like and time-like form factors [15–17]. For $q^2 > 0$, it becomes a dispersion relation between the real and imaginary parts of correlation functions [6], with the following discussions and conclusions remain applicable.

Using the dispersion relation in Eq.(2.3) as a bridge between perturbative and non-perturbative physics, we formulate an inverse problem to calculate non-perturbative quantities from perturbative input. In the deep Euclidean region of $q^2 \ll 0$, the correlation function $\Pi(q^2)$ is perturbatively calculable. Correspondingly, for the spectral function $\text{Im } \Pi(s)$ with $s > 0$, we identify a characteristic scale Λ that separates perturbative and non-perturbative regimes. The region $t_{\min} < s < \Lambda$ contains non-perturbative hadronic information, while for $s > \Lambda$, $\text{Im } \Pi(s)$ can be reliably calculated using perturbation theory. This separation of scales leads to a modified dispersion relation, which we formulate explicitly as an inverse problem:

$$\int_{t_{\min}}^{\Lambda} ds \frac{\text{Im } \Pi(s)}{s - q^2} = \pi \Pi(q^2) - \int_{\Lambda}^{\infty} ds \frac{\text{Im } \Pi(s)}{s - q^2}. \quad (2.4)$$

In this form, the right-hand side serves as a known source term built from perturbative calculations. The task is to solve this equation for the unknown non-perturbative spectral density $\text{Im } \Pi(s)$ in the low-energy region on the left-hand side. The core challenge is to extract the integrand function from information about its integral. Fig. 1 provides a schematic illustration of the inverse problem of dispersion relation.

To simplify the subsequent discussion, we consider a reduced form of the modified dispersion relation:

$$\int_{t_{\min}}^{\Lambda} \frac{f(x)}{x-y} dx = g(y), \quad \text{with } y \in [q_{\text{low}}^2, q_{\text{up}}^2], \quad (2.5)$$

where $\Lambda > t_{\min} \geq 0$ and $q_{\text{low}}^2 < q_{\text{up}}^2 \ll 0$. $f(x)$ is the non-perturbative spectral density $\text{Im } \Pi(s)$, while $g(s)$ represents the perturbatively calculable right-hand side of Eq. (2.4). It is a Fredholm integral equation of the first kind. The corresponding inverse problem is to determine the unknown function $f(x)$ from the known input $g(y)$.

This integral equation can be expressed as an operator equation of the form:

$$Kf = g, \quad \text{with } f \in F, g \in G, \quad (2.6)$$

where $K : F \rightarrow G$ is the linear integral operator, and F and G are the function spaces $F = L^2(t_{\min}, \Lambda)$ and $G = L^2(q_{\text{low}}^2, q_{\text{up}}^2)$, respectively. A brief introduction to L^2 space is given in Appendix A. Consequently, reconstructing f from the operator K and the data function g constitutes a linear inverse problem [22].

3 Ill-posedness of the Inverse Problem

The integral equation Eq. (2.4) provides a framework for obtaining non-perturbative quantities, but solving it presents a substantial challenge. Intuitively, one might discretize Eqs. (2.5) and (2.6) into the matrix formulation $K_{ij}f_j = g_i$, with the formal solution $f_j = (K^{-1})_{ji}g_i$. However, this naive approach is not guaranteed to yield a suitable solution, due to the potential unboundedness (or divergence) of K^{-1} . It is evident from the fact that the inverse of a matrix is proportional to the reciprocal of its determinant $K^{-1} \propto 1/\det(K)$. To illustrate, for Eq. (2.5) with $x \in [0, 1]$ and $y \in [-2, -1]$, the determinant becomes extremely small upon discretization: $\det(K) \approx 10^{-88}$ for 10 points and $\det(K) \approx 10^{-244}$ for 20 points. This implies that any small error in the input $g(y)$ can induce an arbitrarily large deviation in the solution.

The core difficulty is the ill-posed nature of the underlying inverse problem in Eq. (2.5). In this work, it is demonstrated for the first time that while the solution is unique, it is inherently unstable. Consequently, obtaining a reliable numerical solution is highly non-trivial. We begin with a rigorous proof of this ill-posedness in the present section, followed by an examination of resolution strategies in the next section.

An ill-posed problem is defined in contrast to a well-posed one. Mathematically, the notion of well-posedness is formulated in the following way.

Definition 3.1 (well-posedness). Let F and G be normed spaces, $K : F \rightarrow G$ a (linear or nonlinear) mapping. The operator equation $Kf = g$ is called well-posed if the following holds [22]:

1. **Existence:** For every $g \in G$ there is (at least one) $f \in F$ such that $Kf = g$;
2. **Uniqueness:** For every $g \in G$ there is at most one $f \in F$ with $Kf = g$;
3. **Stability:** The solution f depends continuously on g ; that is, for every sequence $(f_n) \subset F$ with $Kf_n \rightarrow Kf$ ($n \rightarrow \infty$), it follows that $f_n \rightarrow f$ ($n \rightarrow \infty$).

A problem is classified as ill-posed if it fails to satisfy any of the aforementioned conditions. The inverse problem of dispersion relation is ill-posed due to the fact of existence, uniqueness, but instability. While a complete mathematical proof of the existence remains open, the physical reality of non-perturbative effects strongly supports the existence of a solution. Before proceeding with the detailed proofs of uniqueness and instability, we first establish the ill-posedness of Eq. (2.5) by analyzing the operator-theoretic structure.

Theorem 3.2. *The operator K in Eqs. (2.5) and (2.6) is linear, bounded, and compact from F to G .*

Proof. It is straightforward to verify that $Kf_1 + Kf_2 = K(f_1 + f_2)$ and $\alpha Kf = K(\alpha f)$ which imply that the $K : F \rightarrow G$ operator is a linear operator.

For any $f \in L^2(t_{\min}, \Lambda)$, applying the Cauchy-Schwarz inequality yields

$$\begin{aligned} \|Kf\|_{L^2(q_{\text{low}}^2, q_{\text{up}}^2)}^2 &= \int_{q_{\text{low}}^2}^{q_{\text{up}}^2} \left(\int_{t_{\min}}^{\Lambda} \frac{1}{x-y} f(x) dx \right)^2 dy \\ &\leq \int_{q_{\text{low}}^2}^{q_{\text{up}}^2} dy \int_{t_{\min}}^{\Lambda} \left(\frac{1}{x-y} \right)^2 dx \int_{t_{\min}}^{\Lambda} f^2(x) dx \\ &= \ln \left| \frac{(\Lambda - q_{\text{up}}^2)(t_{\min} - q_{\text{low}}^2)}{(\Lambda - q_{\text{low}}^2)(t_{\min} - q_{\text{up}}^2)} \right| \|f\|_{L^2(t_{\min}, \Lambda)}^2 < +\infty, \end{aligned} \quad (3.1)$$

where $\ln \left| \frac{(\Lambda - q_{\text{up}}^2)(t_{\min} - q_{\text{low}}^2)}{(\Lambda - q_{\text{low}}^2)(t_{\min} - q_{\text{up}}^2)} \right|$ is a finite constant. From inequality (3.1), it follows that K is a bounded operator.

Define the kernel function $k(y, x) := \frac{1}{x-y}$, $(y, x) \in [q_{\text{low}}^2, q_{\text{up}}^2] \times [t_{\min}, \Lambda]$. Under the disjointness with $q_{\text{up}}^2 < t_{\min}$, k is continuous and bounded on this compact rectangle. In particular

$$\iint_{[q_{\text{low}}^2, q_{\text{up}}^2] \times [t_{\min}, \Lambda]} |k(y, x)|^2 dx dy = C < \infty. \quad (3.2)$$

Thus $k \in L^2([q_{\text{low}}^2, q_{\text{up}}^2] \times [t_{\min}, \Lambda])$, which means that K is a *Hilbert–Schmidt operator* from $L^2(t_{\min}, \Lambda)$ to $L^2(q_{\text{low}}^2, q_{\text{up}}^2)$. A fundamental result in functional analysis states that every Hilbert–Schmidt operator between Hilbert spaces is compact (Hilbert–Schmidt theory). Therefore, K is compact. $\square$

Theorem 3.3. *A compact linear operator $K : F \rightarrow G$ cannot have a bounded inverse unless F has finite dimension.*

This theorem is essential in proving that the inverse problem is ill-posed. In Theorem 3.3, a fundamental result in functional analysis asserts that in an infinite-dimensional Hilbert space, a compact operator cannot have a bounded inverse. Indeed, its inverse is necessarily unbounded and therefore discontinuous. This property immediately implies that the inverse problem for the dispersion relation is ill-posed in the sense of Hadamard. Consequently, the compactness of K lies at the heart of the well-posedness challenges addressed in the subsequent analysis.

In the following, we provide a rigorous proof of both the uniqueness and instability of the inverse problem associated with Eq. (2.5).

Theorem 3.4. (Uniqueness). *Suppose that $f_1(x), f_2(x) \in L^2(t_{\min}, \Lambda)$. If $Kf_1(x) = Kf_2(x) = g(y)$ for all $y \in [q_{\text{low}}^2, q_{\text{up}}^2]$, then $f_1(x) = f_2(x)$, a. e. $x \in [t_{\min}, \Lambda]$.*

Proof. Since K is linear, $Kf_1 - Kf_2 = K(f_1 - f_2) = 0$. Let $\bar{f} = f_1 - f_2$. It suffices to show that $K\bar{f} = 0$ implies $\bar{f}(x) = 0$, for almost every $x \in [t_{\min}, \Lambda]$.

The condition $|q_{\text{up}}^2| > \Lambda$ ensures that $\Pi(s)$ can be perturbatively calculated for $s \in [q_{\text{low}}^2, q_{\text{up}}^2]$. For $x \in [t_{\min}, \Lambda]$ and $y \in [q_{\text{low}}^2, q_{\text{up}}^2]$ with $\Lambda > t_{\min} \geq 0$, $q_{\text{low}}^2 < q_{\text{up}}^2 < 0$, and $|q_{\text{up}}^2| > \Lambda$, we have $|x/y| \leq \Lambda/|q_{\text{up}}^2| < 1$. Therefore, the geometric series expansion $\frac{1}{x-y} = -\frac{1}{y} \sum_{k=0}^{\infty} \left(\frac{x}{y}\right)^k$ converges absolutely. Substituting into $K\bar{f}$ gives

$$K\bar{f} = \int_{t_{\min}}^{\Lambda} \frac{1}{x-y} \bar{f}(x) dx = -\frac{1}{y} \int_{t_{\min}}^{\Lambda} \sum_{k=0}^{\infty} \left(\frac{x}{y}\right)^k \bar{f}(x) dx. \quad (3.3)$$

Since $\left| \sum_{k=0}^{\infty} \left(\frac{x}{y}\right)^k \bar{f}(x) \right| \leq \sum_{k=0}^{\infty} \left| \frac{\Lambda}{q_{\text{up}}^2} \right|^k |\bar{f}(x)| = \frac{1}{1-\Lambda/|q_{\text{up}}^2|} |\bar{f}(x)|$ and $\int_{t_{\min}}^{\Lambda} |\bar{f}(x)| dx < +\infty$ (because $\bar{f} \in L^2 \subset L^1$ on a bounded interval), the dominated convergence theorem allows exchanging the sum and integral.

Hence,

$$-y \int_{t_{\min}}^{\Lambda} \frac{1}{x-y} \bar{f}(x) dx = \sum_{k=0}^{\infty} \frac{1}{y^k} \int_{t_{\min}}^{\Lambda} x^k \bar{f}(x) dx. \quad (3.4)$$

Using $K\bar{f} = 0$, we have

$$\sum_{k=0}^{\infty} \frac{1}{y^k} \int_{t_{\min}}^{\Lambda} x^k \bar{f}(x) dx = 0, \quad y \in [q_{\text{low}}^2, q_{\text{up}}^2]. \quad (3.5)$$

The above equation is an infinite sum. As a necessary step toward the conclusion $\bar{f}(x) = 0$, we should firstly establish that all moments vanish:

$$m_k := \int_{t_{\min}}^{\Lambda} x^k \bar{f}(x) dx = 0, \quad k = 0, 1, 2, \dots \quad (3.6)$$

The proof proceeds by considering two cases: q_{low}^2 being negative infinity or finite.

- Case 1: $q_{\text{low}}^2 = -\infty$. From Eq. (3.5), we have

$$\sum_{k=0}^{\infty} \frac{m_k}{y^k} = 0, \quad y \in (-\infty, q_{\text{up}}^2). \quad (3.7)$$

Taking the limit $y \rightarrow -\infty$ gives the first vanishing moment, $m_0 = 0$. Multiplying by y and taking $y \rightarrow -\infty$ again yields the second $m_1 = 0$. Proceeding inductively, we obtain $m_k = 0$ for all k .

- Case 2: q_{low}^2 is finite, i.e., $q_{\text{low}}^2 > -\infty$. Let $z \in \mathbb{C}$ with $|z| \geq |q_{\text{up}}^2|$. Then

$$\left| \frac{m_k}{z^k} \right| \leq \frac{|m_k|}{|q_{\text{up}}^2|^k} \leq \left(\frac{\Lambda}{|q_{\text{up}}^2|} \right)^k \|\tilde{f}\|_{L^1}. \quad (3.8)$$

By the Weierstrass M-test, the series $\sum_{k=0}^{\infty} m_k/z^k$ converges absolutely and uniformly on $D = \{z : |z| \geq |q_{\text{up}}^2|\}$. Since each term m_k/z^k is analytic on D , the sum $\sum_{k=0}^{\infty} m_k/z^k$ is also analytic on D by the Weierstrass convergence theorem. From Eq. (3.5), $\sum_{k=0}^{\infty} m_k/y^k = 0$ holds for $y \in [q_{\text{low}}^2, q_{\text{up}}^2]$. By analytic continuation, it holds for all $y \in (-\infty, q_{\text{up}}^2]$. The limiting argument from Case 1 then gives $m_k = 0$ for all k .

In both cases, all moments of $\tilde{f}(x)$ vanish. Since $C[t_{\min}, \Lambda]$ is dense in $L^2(t_{\min}, \Lambda)$, for any $\epsilon > 0$, there exists $\tilde{f}(x) \in C[t_{\min}, \Lambda]$ such that $\|\tilde{f}(x) - \tilde{f}(x)\|_{L^2} \leq \epsilon$. By the Weierstrass approximate theorem, there exists a polynomial $Q_n(x)$ such that $\|\tilde{f}(x) - Q_n(x)\|_{C[t_{\min}, \Lambda]} \leq \epsilon$. Hence,

$$\begin{aligned} \|\tilde{f}(x) - Q_n(x)\|_{L^2} &\leq \|\tilde{f}(x) - \tilde{f}(x)\|_{L^2} + \|\tilde{f}(x) - Q_n(x)\|_{L^2} \\ &\leq \epsilon + \sqrt{b-a} \|\tilde{f}(x) - Q_n(x)\|_{C[t_{\min}, \Lambda]} \\ &\leq \epsilon + \epsilon \sqrt{\Lambda - t_{\min}}. \end{aligned} \quad (3.9)$$

Since all moments vanish, $\int_{t_{\min}}^{\Lambda} \tilde{f}(x) Q_n(x) dx = 0$. Applying the Cauchy-Schwarz inequality,

$$\begin{aligned} \|\tilde{f}(x)\|_{L^2}^2 &= \int_{t_{\min}}^{\Lambda} \tilde{f}^2(x) dx = \int_{t_{\min}}^{\Lambda} (\tilde{f}^2(x) - \tilde{f}(x) Q_n(x)) dx \\ &\leq \left(\int_{t_{\min}}^{\Lambda} \tilde{f}^2(x) dx \right)^{\frac{1}{2}} \left(\int_{t_{\min}}^{\Lambda} |\tilde{f}(x) - Q_n(x)|^2 dx \right)^{\frac{1}{2}} \\ &= \|\tilde{f}(x)\|_{L^2} \cdot \|\tilde{f}(x) - Q_n(x)\|_{L^2} \\ &\leq (\epsilon + \epsilon \sqrt{\Lambda - t_{\min}}) \|\tilde{f}(x)\|_{L^2}. \end{aligned} \quad (3.10)$$

Thus, $\|\tilde{f}(x)\|_{L^2} \leq \epsilon + \epsilon \sqrt{\Lambda - t_{\min}}$. Letting $\epsilon \rightarrow 0$, we obtain $\|\tilde{f}(x)\|_{L^2} = 0$, i. e. $\tilde{f}(x) = 0$, a.e., $x \in [t_{\min}, \Lambda]$. Therefore, $f_1(x) = f_2(x)$ almost everywhere, proving uniqueness.

The above proof relies on the geometric series expansion and the moment argument for $y \in [q_{\text{low}}^2, q_{\text{up}}^2]$ with $|q_{\text{up}}^2| > \Lambda$. Remarkably, the uniqueness result remains valid even when $|q_{\text{up}}^2| < \Lambda$, provided that the interval $[q_{\text{low}}^2, q_{\text{up}}^2]$ is non-degenerate. This is a consequence of the analyticity of the integral $\int_{t_{\min}}^{\Lambda} \frac{\tilde{f}(x)}{x-y} dx$ as a function of y , where y is a complex number not located in the interval $[t_{\min}, \Lambda]$ on the real axis. If this function vanishes on any subinterval of $(-\infty, 0)$, then by the identity theorem for analytic functions, it vanishes identically on $(-\infty, t_{\min})$. Hence, $\tilde{f}(x) = 0$, and therefore $f_1(x) = f_2(x)$ almost everywhere in

$[t_{\min}, \Lambda]$. Note that the above argument regarding analyticity of the integral also ensures the uniqueness of the inverse problem for dispersion relation when $y \in [q_{\text{low}}^2, q_{\text{up}}^2]$ with $0 < \Lambda < q_{\text{low}}^2 < q_{\text{up}}^2$ [6].

This completes the proof. $\square$

Theorem 3.5. (Instability.) *Consider the inverse problem in Eq. (2.5) with $0 < t_{\min} < \Lambda < +\infty$ and $0 > q_{\text{up}}^2 > q_{\text{low}}^2 > -\infty$. Then the problem is unstable: there exists a sequence of perturbed data $g_n \rightarrow g$ in $L^2(q_{\text{low}}^2, q_{\text{up}}^2)$, but the corresponding solutions f_n satisfy $\|f_n - f\|_{L^2(t_{\min}, \Lambda)} \rightarrow \infty$.*

Proof. We construct a specific counterexample. Let $t_{\min} = 0$ and $\Lambda = 2$, and define the perturbed solution

$$f_n(x) = f(x) + \sqrt{n} \cos(n\pi x), \quad x \in [0, 2]. \quad (3.11)$$

Let $g_n(y)$ be the data corresponding to f_n , i.e.,

$$g_n(y) - g(y) = \int_0^2 \frac{\sqrt{n} \cos(n\pi x)}{x - y} dx. \quad (3.12)$$

Since $y \leq q_{\text{up}}^2 < 0$ and $x \in [0, 2]$, we have $x - y \geq -y \geq -q_{\text{up}}^2 =: d > 0$. Thus the kernel is smooth and bounded. Integrating it by parts:

$$g_n(y) - g(y) = \frac{\sqrt{n}}{n\pi} \int_0^2 \frac{1}{x - y} d(\sin(n\pi x)) = \frac{1}{\sqrt{n}\pi} \left[\frac{\sin(n\pi x)}{x - y} \Big|_{x=0}^{x=2} + \int_0^2 \frac{\sin(n\pi x)}{(x - y)^2} dx \right]. \quad (3.13)$$

Since $\sin(2n\pi) = \sin(0) = 0$, the boundary term vanishes. Using $|\sin(n\pi x)| \leq 1$ and $|x - y| \geq d = -q_{\text{up}}^2$, we obtain $|g_n(y) - g(y)| \leq \frac{1}{\sqrt{n}\pi} \frac{2}{d^2}$. Therefore,

$$\|g_n - g\|_{L^2(q_{\text{low}}^2, q_{\text{up}}^2)} \leq \frac{2}{\sqrt{n}\pi d^2} \sqrt{q_{\text{up}}^2 - q_{\text{low}}^2} =: \frac{M}{\sqrt{n}} \xrightarrow{n \rightarrow \infty} 0, \quad (3.14)$$

where $M = 2 \sqrt{q_{\text{up}}^2 - q_{\text{low}}^2} / (\pi d^2)$ is finite and independent of n .

On the other hand, the error in the solution is

$$\|f_n - f\|_{L^2(0,2)} = \left(\int_0^2 n \cos^2(n\pi x) dx \right)^{1/2} = \sqrt{n} \left(\int_0^2 \cos^2(n\pi x) dx \right)^{1/2} = \sqrt{n} \xrightarrow{n \rightarrow \infty} \infty \quad (3.15)$$

since $\int_0^2 \cos^2(n\pi x) dx = 1$ for all integer $n \geq 1$.

Thus, an arbitrarily small perturbation in the data can produce an arbitrarily large error in the solution, proving the instability of the inverse problem. $\square$

In summary, we have rigorously established that the inverse problem associated with the dispersion relation is ill-posed: although the solution is unique, it lacks stability. This ill-posedness is illustrated in Fig. 2. Uniqueness ensures a one-to-one mapping between any perturbative input and its corresponding non-perturbative solution. However, the absence of stability poses a fundamental challenge: arbitrarily small errors in the input data can lead to large deviations in the reconstructed solution. Critically, because perturbation theory is only asymptotically convergent, such input errors are inevitable in practice. Consequently, regularization techniques are essential to stabilize the inversion procedure.

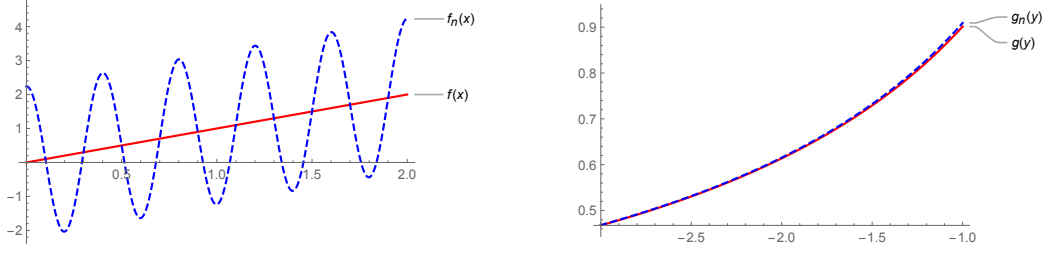


Figure 2: Illustration of uniqueness but instability in the inverse problem. The functions $f(x) = x$ and $f_n(x) = x + \sqrt{5} \cos(5\pi x)$ yield nearly identical input data $g(y)$ and $g_n(y)$, yet differ significantly in the solution space.

4 Regularization Methods

In this section, we discuss the application of regularization methods to solve ill-posed inverse problems. Regularization theory has been rigorously developed in mathematics since the mid-20th century and is extensively covered in numerous texts [20–23]. Here, we provide a brief introduction to ensure the logical flow of the article and to make the material accessible for readers with a physics background. The basic idea of the regularization method is illustrated in Fig. 3

4.1 General Regularization Theory

The basic idea of regularization methods is to approximate an ill-posed problem by a well-posed problem that is close to the original one, ensuring that the approximate solution converges to the true solution as the input error diminishes.

Definition 4.1 (Regularization strategy). A regularization strategy is a family of linear and bounded operators $R_\alpha : G \rightarrow F$, $\alpha > 0$, such that

$$\lim_{\alpha \rightarrow 0} R_\alpha K f = f \quad \text{for all } f \in F, \quad (4.1)$$

where α is called the regularization parameter.

From this definition, it follows that the regularization operators possess two fundamental properties. Firstly, the operator R_α is bounded for any fixed $\alpha > 0$. This ensures that the approximate solution $f_\alpha = R_\alpha g$ depends continuously on the data g , rendering the approximate problem well-posed and numerically solvable even in the presence of data perturbations. Secondly, as $\alpha \rightarrow 0$, the composite operators $R_\alpha K$ converge pointwise to the identity operator on F , implying $R_\alpha g \rightarrow K^{-1}g$. The limit establishes both convergence and consistency.

Regularization theory for ill-posed problems can be intuitively understood through a linear algebra analogy. Consider a square matrix A in a finite-dimensional space, which is positive semidefinite and

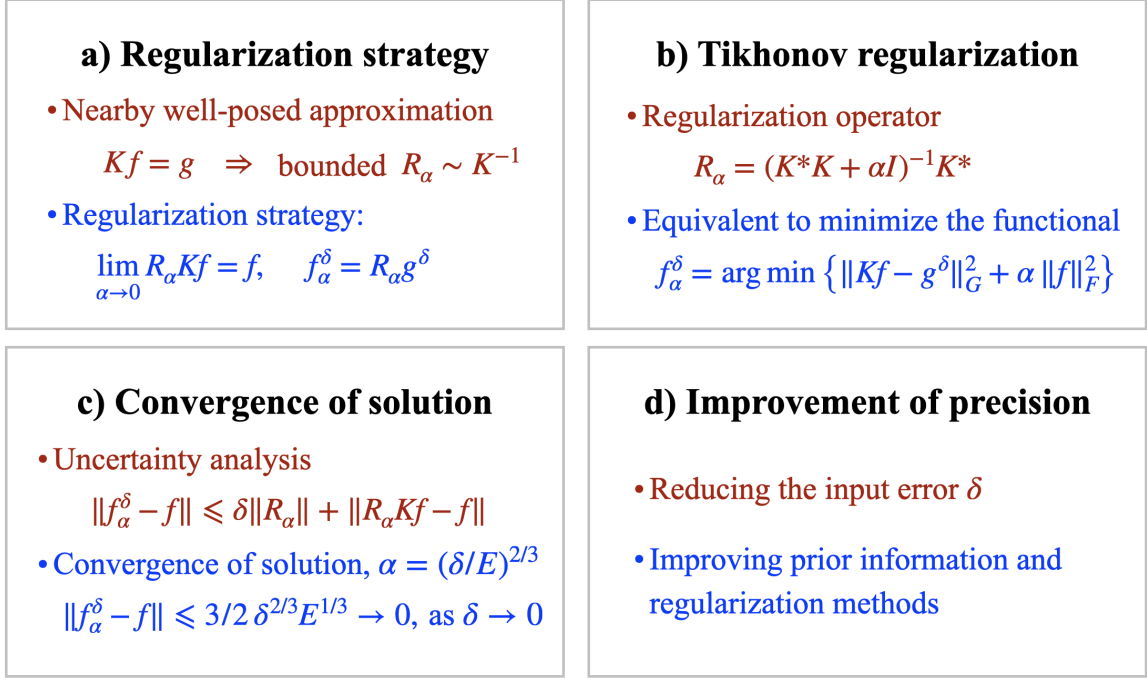


Figure 3: Illustration of the regularization method.

singular (non-invertible). A common regularization method approximates A by $A + \alpha I$, where I is the identity matrix, and $\alpha > 0$. This approximation is consistent, since $A + \alpha I \rightarrow A$ as $\alpha \rightarrow 0$. In this context, the regularization operator corresponds to

$$R_\alpha = (A + \alpha I)^{-1}. \quad (4.2)$$

In the following, we demonstrate that R_α is well-defined and bounded. For matrices, non-invertibility is equivalent to a vanishing determinant, i.e., $\det(A) = 0$. Since the determinant equals the product of the eigenvalues λ_i (i.e., $\det(A) = \prod_i \lambda_i$), $\det(A) = 0$ implies that at least one eigenvalue λ_i is zero. Regarding the eigenvalue equation $Ax = \lambda x$, we observe that $(A + \alpha I)x = (\lambda + \alpha)x$. Thus, $\lambda_i + \alpha$ are the eigenvalues of $A + \alpha I$. Since A is positive semidefinite, $\lambda_i \geq 0$. For any $\alpha > 0$, the shifted eigenvalues satisfy $\lambda_i + \alpha \neq 0$. Consequently, the determinant becomes $\det(A + \alpha I) = \prod_i (\lambda_i + \alpha) \neq 0$. This implies that $A + \alpha I$ is invertible. Therefore, the regularization operator $R_\alpha = (A + \alpha I)^{-1}$ exists and is bounded. This illustrates that adding the term αI removes the singularity and renders the problem well-posed.

In the following, we discuss the error estimation. In practice, input data g is typically observed as noisy data $g^\delta \in G$ with $\|g^\delta - g\| \leq \delta$. We define

$$f_\alpha^\delta := R_\alpha g^\delta \quad (4.3)$$

as an approximation of the true solution f of $Kf = g$. The error of the solution is quantified by the norm of the discrepancy, $\|f_\alpha^\delta - f\|$. By invoking the triangle inequality, this error can be decomposed into two

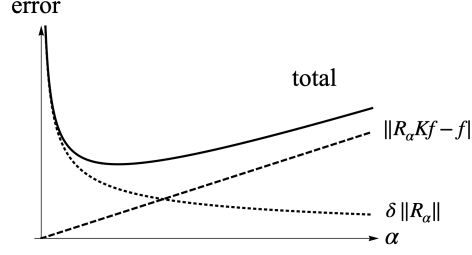


Figure 4: Illustration of the total error and its decomposition into two components from $\|R_\alpha K f - f\|$ and $\delta \|R_\alpha\|$.

parts:

$$\begin{aligned}
 \|f_\alpha^\delta - f\| &\leq \|f_\alpha^\delta - f_\alpha\| + \|f_\alpha - f\| \\
 &= \|R_\alpha g^\delta - R_\alpha g\| + \|R_\alpha g - f\| \\
 &\leq \|R_\alpha\| \|g^\delta - g\| + \|R_\alpha K f - f\|.
 \end{aligned} \tag{4.4}$$

Here, the final step relies on the inequality property of the operator norm. The formal definition of the operator norm and the associated inequality theorem are provided in definition A.4 and theorem A.5 of Appendix A, respectively. Given the error of noisy input data $\|g^\delta - g\| \leq \delta$, we obtain the key error bound:

$$\|f_\alpha^\delta - f\| \leq \|R_\alpha K f - f\| + \delta \|R_\alpha\|. \tag{4.5}$$

This relation is central to our uncertainty analysis.

The two terms in Eq. (4.5) are analyzed below, with their impact on the total error depicted in Fig. 4. The first term, $\|R_\alpha K f - f\|$, quantifies the error due to the regularization approximation. According to the definition of the regularization operator in Eq. (4.1), this term vanishes as $\alpha \rightarrow 0$ (i.e., $\|R_\alpha K f - f\| \rightarrow 0$). Thus, the approximation error decreases as α diminishes and increases as α grows. The second term, $\delta \|R_\alpha\|$, represents the propagation of the data error into the solution error, scaled by the operator norm $\|R_\alpha\|$. For ill-posed problems where K^{-1} is unbounded, the regularized operator R_α approaches K^{-1} as $\alpha \rightarrow 0$, causing the operator norm $\|R_\alpha\|$ to blow up to infinity. In this sense, $\|R_\alpha\|$ increases as α decreases (increasing ill-conditioning) and decreases as α increases (improving stability).

As illustrated in Fig. 4, the regularization parameter α cannot be chosen arbitrarily large or small. Rather, it must strike a balance between two competing requirements. On the one hand, ensuring the convergence of the regularized solution to the true solution, $\|(R_\alpha - K^{-1})g\| \rightarrow 0$, calls for a sufficiently small α . On the other hand, maintaining the stability of the solution requires that the operator norm $\|R_\alpha\|$ remain bounded, which prohibits α from being too small. Consequently, the choice of α reflects an inherent trade-off between approximation accuracy and numerical stability.

Therefore, the fundamental principle of regularization for solving ill-posed problems is to establish a strategy for selecting the parameter $\alpha = \alpha(\delta)$. This strategy aims to balance the approximation error and

the stability term by effectively minimizing the following total error bound with respect to α :

$$\|R_\alpha Kf - f\| + \delta \|R_\alpha\|.$$

Crucially, as the noise level $\delta \rightarrow 0$, the parameter choice must ensure that this total error converges to zero. This convergence property constitutes a core tenet of regularization theory: regularized solutions must converge to the true solution as data accuracy improves.

Definition 4.2 (Admissible regularization strategy). A regularization strategy $\alpha = \alpha(\delta)$ is called *admissible* if, as the noise level $\delta \rightarrow 0$, the parameter satisfies $\alpha(\delta) \rightarrow 0$ and the regularized solution converges to the true solution¹, $f_\alpha^\delta \rightarrow f$.

An intuitive interpretation of this definition involves two key conditions. First, the regularization parameter must satisfy $\alpha(\delta) \rightarrow 0$ as $\delta \rightarrow 0$, which is easily achieved by choosing α as a decaying function of the noise level. Second, the total error bound $\|R_\alpha Kf - f\| + \delta \|R_\alpha\|$ must vanish in this limit. The crucial challenge lies in the stability term $\delta \|R_\alpha\|$. Since $\|R_\alpha\|$ typically diverges as $\alpha \rightarrow 0$ (an inverse relationship), the parameter choice strategy must ensure that this divergence is sufficiently slow. Specifically, $\|R_{\alpha(\delta)}\|$ must grow strictly slower than $1/\delta$; only then will the product $\delta \|R_\alpha\|$ converge to zero, guaranteeing the convergence of the regularized solution. In the following subsection, we will demonstrate the convergence of approximate solutions for Tikhonov regularization.

It is important to clarify that our discussion does not imply the notion of “fine-tuning” often encountered in physics. While the approximate solution formally depends on the parameter α , the toy models presented in the next section demonstrate that the solutions remain insensitive to α over a broad range of values. This behavior aligns with physical expectations: a valid regularized solution should exhibit a stable plateau with respect to variations in α .

4.2 Tikhonov Regularization

Before introducing a specific regularization method, we recall that such methods are designed to satisfy two fundamental properties: (1) restoring well-posedness by ensuring the existence, uniqueness, and stability of a solution; and (2) guaranteeing convergence, meaning that the regularized solution approaches the true solution as the error in the input data tends to zero.

In this work, we employ the Tikhonov regularization method, one of the most prevalent and well-established techniques in the field. Originally introduced by A. N. Tikhonov in the 1940s [20, 21], this

¹Formally, this definition requires that as $\delta \rightarrow 0$, it satisfies $\alpha(\delta) \rightarrow 0$ and

$$\sup \{ \|R_{\alpha(\delta)} g^\delta - f\| : g^\delta \in G, \|Kf - g^\delta\| \leq \delta \} \rightarrow 0.$$

approach has become pervasive, underpinning the development of methodologies for a vast array of ill-posed problems. Distinguished by its conceptual simplicity, the method offers an elegant framework that is both easy to implement and theoretically rigorous. In this subsection, we first introduce the Tikhonov regularization operator, then demonstrate the convergence of its solution, and finally present the minimization of the Tikhonov functional for practical applications.

Tikhonov Regularization Operator

Analogous to the matrix example in the previous subsection and Eq. (4.2), Tikhonov regularization involves adding a term αI . However, unlike the case of a square matrix, the operator $K : F \rightarrow G$ in the equation $Kf = g$ maps between two generally distinct Hilbert spaces. Directly adding αI to K is ill-defined, since the identity operator maps F onto itself, resulting in a dimensional mismatch. To resolve this, applying the adjoint operator K^* (see Definition A.6) to both sides of $Kf = g$ yields the normal equation $K^*Kf = K^*g$. The composite operator $K^*K : F \rightarrow F$ is self-adjoint, making it analogous to a square matrix. The original ill-posed problem, formally corresponding to $f = (K^*K)^{-1}K^*g$, is transformed into a well-posed approximate problem by introducing the regularization term αI . The regularized solution is given by

$$f_\alpha = (K^*K + \alpha I)^{-1}K^*g.$$

This expression defines the Tikhonov regularization operator

$$R_\alpha = (K^*K + \alpha I)^{-1}K^*. \quad (4.6)$$

For noisy data g^δ , the regularized solution is computed as $f_\alpha^\delta = (K^*K + \alpha I)^{-1}K^*g^\delta$. The operator R_α in Eq. (4.6) provides a stable and bounded approximation of the unbounded inverse of K , and satisfies the essential convergence property in Eq. (4.1). Hence, R_α constitutes a valid regularization operator.

Convergence of solution

In the following, we discuss the convergence of solution under Tikhonov regularization. According to Definition 4.2, the convergence of solutions implies that as $\delta \rightarrow 0$, we have $\alpha(\delta) \rightarrow 0$ and $\|f_\alpha^\delta - f\| \rightarrow 0$. That is, as the precision of the input data improves, the approximate solutions converge to the true solution.

Note that this should not be confused with the convergence of regularization operators, which refers to the fact that as $\alpha \rightarrow 0$, $\|(R_\alpha - K^{-1})g\| \rightarrow 0$, i.e., $R_\alpha \rightarrow K^{-1}$. In this case, the ill-posedness is recovered. What is meaningful is that the regularization operator continues to function as a regularization. By first letting $\delta \rightarrow 0$ and consequently $\alpha(\delta) \rightarrow 0$, we ensure $\|f_\alpha^\delta - f\| \rightarrow 0$. This is precisely the convergence of solutions.

The convergence of solutions followed directly from the error bound given in Eq. (4.5):

$$\|f_\alpha^\delta - f\| \leq \|R_\alpha K f - f\| + \delta \|R_\alpha\|.$$

To prove convergence, we derive upper bounds for the approximate error $\|R_\alpha K f - f\|$ and the propagated data error $\delta \|R_\alpha\|$, respectively. We then demonstrate that the sum of these terms vanishes as $\delta \rightarrow 0$, provided that the regularization parameter $\alpha(\delta)$ is chosen appropriately.

First, we analyze the approximation error $\|R_\alpha K f - f\|$. Consider a specific *a priori* source condition: $f = K^* K z$ for some $z \in F$ with $\|z\|_F \leq E$. This condition imposes both boundedness and smoothness on the true solution f . The boundedness requirement is naturally satisfied in physical contexts, as all physical quantities are inherently finite. Regarding smoothness, the representation $f = K^* K z$ implies that f lies in the range of the composite operator $K^* K$. Since compact operators (and their adjoints) possess a smoothing effect, applying $K^* K$ to z yields a solution f that is smoother than the source element z . Such smoothness assumptions are standard in most inverse problems. In the following, we investigate the behavior of the approximation error $\|R_\alpha K f - f\|$ as a function of the regularization parameter α .

Using the definition in Eq. (4.6) and letting $T = K^* K$, the error term simplifies as follows:

$$\begin{aligned} R_\alpha K f - f &= (T + \alpha I)^{-1} T f - f \\ &= (T + \alpha I)^{-1} (T f - (T + \alpha I) f) \\ &= -\alpha (T + \alpha I)^{-1} f \\ &= -\alpha (T + \alpha I)^{-1} T z, \end{aligned}$$

where the last step utilizes the source condition $f = T z$. Taking the norm of both sides and applying the compatibility property of the operator norm (see Theorem A.5), we obtain the following estimate

$$\begin{aligned} \|R_\alpha K f - f\| &= \alpha \|(T + \alpha I)^{-1} T z\| \\ &\leq \alpha \|(T + \alpha I)^{-1} T\| \cdot \|z\|_F \\ &\leq \alpha E \|(T + \alpha I)^{-1} T\|, \end{aligned} \tag{4.7}$$

where we have employed the *a priori* bound $\|z\|_F \leq E$.

Let $A = (T + \alpha I)^{-1} T$. We can rewrite A as follows $A = (T + \alpha I)^{-1} (T + \alpha I - \alpha I) = I - \alpha (T + \alpha I)^{-1}$. Since $T = K^* K$ is a self-adjoint operator, the resolvent $(T + \alpha I)^{-1}$ is also self-adjoint; consequently, A is self-adjoint. According to the definition of the norm for a self-adjoint operator, we have $\|A\| = \sup_{\|f\|=1} |(A f, f)_F|$. For any unit vector f (i.e., $\|f\| = 1$), the inner product expands to:

$$\begin{aligned} (A f, f) &= (f, f) - \alpha ((T + \alpha I)^{-1} f, f) \\ &= 1 - \alpha ((T + \alpha I)^{-1} f, f). \end{aligned} \tag{4.8}$$

Observe that $(Tf, f) = (K^*Kf, f) = (Kf, Kf) = \|Kf\|^2 \geq 0$. Thus, T is a positive semi-definite operator ($T \geq 0$). It follows that $T + \alpha I \geq \alpha I$. By the monotonicity property of the inverse for positive definite operators, we obtain $(T + \alpha I)^{-1} \leq (\alpha I)^{-1} = \frac{1}{\alpha} I$. This implies that for any unit vector f , $0 \leq ((T + \alpha I)^{-1}f, f) \leq \frac{1}{\alpha}(f, f) = \frac{1}{\alpha}\|f\|^2 = \frac{1}{\alpha}$. Multiplying by α , we get $0 \leq \alpha((T + \alpha I)^{-1}f, f) \leq 1$. Substituting this bound into Eq. (4.8) yields $0 \leq (Af, f) \leq 1$. Therefore, the operator norm satisfies $\|A\| \leq 1$. Applying this result to Eq. (4.7), we conclude:

$$\|R_\alpha Kf - f\| \leq \alpha E. \quad (4.9)$$

Under different a priori source conditions, the results may differ slightly. For instance, consider the source condition $f = K^*z$ with $z \in G$ and $\|z\|_G \leq E$. In this case, the error bound becomes $\|R_\alpha Kf - f\| \leq \frac{\sqrt{\alpha}}{2}E$. Compared to the case where $f = K^*Kz$, the regularity (smoothness) of f is weaker. Consequently, the convergence rate of $\|R_\alpha Kf - f\|$ with respect to α is slower, specifically $O(\sqrt{\alpha})$ versus $O(\alpha)$. In the following discussion, we focus exclusively on the case where $f = K^*Kz$.

Next, we analyze the term $\delta \|R_\alpha\|$. Since the regularized solution is given by $f_\alpha^\delta = R_\alpha g^\delta$, we invoke Definition A.8 for the operator norm: $\|R_\alpha\| = \sup_{g^\delta \neq 0} \frac{\|R_\alpha g^\delta\|_F}{\|g^\delta\|_G}$. Consequently, estimating $\|R_\alpha\|$ is equivalent to finding an upper bound for the ratio of $\|R_\alpha g^\delta\|$ to $\|g^\delta\|$.

The regularized solution satisfies the normal equation $(K^*K + \alpha I)f_\alpha^\delta = K^*g^\delta$. Taking the inner product of both sides with f_α^δ yields $((K^*K + \alpha I)f_\alpha^\delta, f_\alpha^\delta) = (K^*g^\delta, f_\alpha^\delta)$, which expands to $\|Kf_\alpha^\delta\|^2 + \alpha\|f_\alpha^\delta\|^2 = (g^\delta, Kf_\alpha^\delta)$. By the Cauchy-Schwarz inequality, the right-hand side is bounded by $(g^\delta, Kf_\alpha^\delta) \leq \|g^\delta\| \cdot \|Kf_\alpha^\delta\|$. Meanwhile, applying the Arithmetic Mean-Geometric Mean (AM-GM) inequality to the left-hand side gives $\|Kf_\alpha^\delta\|^2 + \alpha\|f_\alpha^\delta\|^2 \geq 2\sqrt{\alpha}\|Kf_\alpha^\delta\| \cdot \|f_\alpha^\delta\|$. Combining these results leads to $2\sqrt{\alpha}\|Kf_\alpha^\delta\| \cdot \|f_\alpha^\delta\| \leq \|g^\delta\| \cdot \|Kf_\alpha^\delta\|$. Assuming $\|Kf_\alpha^\delta\| \neq 0$ (the case where it equals zero holds trivially), we divide by $\|Kf_\alpha^\delta\|$ to obtain $\frac{\|f_\alpha^\delta\|}{\|g^\delta\|} \leq \frac{1}{2\sqrt{\alpha}}$. Since $f_\alpha^\delta = R_\alpha g^\delta$, taking the supremum over all $g^\delta \neq 0$ allows us to conclude that the operator norm satisfies:

$$\|R_\alpha\| \leq \frac{1}{2\sqrt{\alpha}}. \quad (4.10)$$

This result aligns with the discussion in the previous subsection: as α decreases, $\|R_\alpha\|$ increases and may even diverge; conversely, as α increases, $\|R_\alpha\|$ decreases, as depicted in Fig. 4. Consequently, we establish the final bound for the noise propagation term:

$$\delta \|R_\alpha\| \leq \frac{\delta}{2\sqrt{\alpha}}. \quad (4.11)$$

The error bounds in Eqs. (4.9) and (4.11) can also be derived using the spectral theorem and singular value decomposition. However, these derivations are omitted here for brevity.

By combining the error bounds from Eqs. (4.9) and (4.11), we establish the convergence of the

solution by minimizing the upper bound

$$\|f_\alpha^\delta - f\| \leq \alpha E + \frac{\delta}{2\sqrt{\alpha}}. \quad (4.12)$$

The minimum is achieved by choosing $\alpha = (\delta/E)^{2/3}$, which yields

$$\|f_\alpha^\delta - f\| \leq \frac{3}{2} \delta^{\frac{2}{3}} E^{\frac{1}{3}}. \quad (4.13)$$

It is evident that, regardless of the magnitude of E , the solution error converges to zero,

$$\|f_\alpha^\delta - f\| \rightarrow 0, \quad \delta \rightarrow 0. \quad (4.14)$$

This demonstrates the convergence of the solution: under Tikhonov regularization, the approximate regularized solution converges to the true solution in the limit of vanishing noise $\delta \rightarrow 0$.

The convergence of the solution is the most critical property of regularization methods. It implies that the inverse problem approach can be systematically improved as the input error decreases. Conversely, even when the noise level δ is not infinitesimally small, Eq. (4.13) demonstrates that the uncertainty of the solution possesses an upper bound. This indicates that the uncertainty remains under control, thereby confirming that regularization ensures stability. Furthermore, in establishing solution convergence, the choice of the regularization parameter α is not arbitrary. It must depend on δ . For a fixed δ , α must be chosen carefully. It cannot be too large nor too small. This aligns with the discussion in the previous subsection. Finally, although the *a priori* bound E may be difficult to estimate in practice, this does not hinder the convergence of the solution. Regardless of the specific method used to select α , the convergence of the solution is guaranteed.

Minimization of the Tikhonov functional

The discussions thus far have been grounded in the regularization operator R_α and the corresponding operator equation $f_\alpha^\delta = R_\alpha g^\delta$. Alternatively, this can be viewed through the lens of minimizing the Tikhonov functional, an equivalent scheme that is often preferred in practical applications. This variational approach offers a new perspective on the regularization of ill-posed problems. Furthermore, it provides a versatile foundation for improving regularization methods—for instance, by modifying norm spaces to reflect *a priori* knowledge.

Theorem 4.3. *Let $K : F \rightarrow G$ be a linear and bounded operator between Hilbert spaces and $\alpha > 0$. Then the Tikhonov functional J_α ,*

$$J_\alpha(f) := \|Kf - g^\delta\|_G^2 + \alpha \|f\|_F^2 \quad \text{for } f \in F, \quad (4.15)$$

has a unique minimizer

$$f_\alpha^\delta = \arg \min_{f \in F} J_\alpha(f). \quad (4.16)$$

This minimizer f_α^δ is the unique solution of the normal equation

$$\alpha f_\alpha^\delta + K^* K f_\alpha^\delta = K^* g^\delta.$$

The proof of the existence of the minimizer and its equivalence to the solution of the normal equation is shown in Appendix B.

The minimization of the Tikhonov functional offers an intuitive perspective on regularization. The first term, $\|Kf - g^\delta\|_G^2$, represents the data fidelity or residual, analogous to χ^2 in fitting procedures. While minimizing the residual term seeks a solution to $Kf = g^\delta$, driving it too close to zero often leads to overfitting the noise in g^δ . This instability arises because K is typically a continuous smoothing operator that suppresses high-frequency variations. In a discretized setting, this implies that a solution f with rapidly alternating signs (where adjacent components f_i and f_{i+1} cancel each other out) can still yield a product Kf extremely close to the observed data g^δ . Consequently, minimizing the residual term alone fails to penalize such high-frequency oscillations, resulting in an unstable and physically meaningless solution.

To counteract this, the second term, $\alpha \|f\|_F^2$, serves as a regularization or penalty term that encodes prior assumptions about the smoothness or boundedness of the solution. Mathematically, since high-frequency oscillations typically correspond to large norms in the function space F , penalizing $\|f\|_F^2$ effectively suppresses these unstable components that the smoothing operator K overlooks. The regularization parameter $\alpha > 0$ acts as a crucial balance lever, mediating the trade-off between the fidelity of the data and the stability of the solution. By preventing the norm from growing excessively, a properly chosen α mitigates the ill-posedness of the original problem. As established in Theorem 4.3, this combined functional is strictly convex, guaranteeing a unique and stable regularized solution f_α^δ that converges to the true solution as the noise level vanishes.

A key strength of the Tikhonov functional is its adaptability to a priori physical conditions. Specifically, the penalty term can be tailored to reflect the known properties of the solution. For instance, while the standard L^2 norm ($\alpha \|f\|_{L^2}^2$) penalizes the magnitude of the solution, one may substitute it with the H^1 norm ($\alpha \|f\|_{H^1}^2$) if smoothness is expected. As detailed in Appendix A, the H^1 norm incorporates derivative information to explicitly promote smoothness, and Section 5 demonstrates the efficacy of this modification on several toy models.

However, a significant limitation inherent to standard Tikhonov regularization is the tendency to induce over-smoothing. Acting as a low-pass filter, the penalty term uniformly suppresses high-frequency

components. While this effectively stabilizes the inversion against noise, it often excessively dampens sharp features. Consequently, if the true physical spectral function contains narrow resonance peaks, discontinuities, or sparse structures, the reconstructed solution may appear artificially broadened and blurred. This results in a loss of resolution and a systematic underestimation of peak amplitudes.

During the past half-century, numerous advanced regularization techniques have evolved to address the limitations of standard methods, such as iterative regularization, discrete regularization, Bayesian regularization, and so on [22, 23]. However, in this work, which establishes the initial mathematical foundation for an inverse problem approach to non-perturbative QCD, we focus exclusively on the Tikhonov framework. As the earliest and most fundamental method for solving ill-posed problems, Tikhonov regularization is distinguished by its conceptual clarity and computational efficiency, making it an ideal starting point. Future studies will build on this basis by exploring more sophisticated methods to address complex physical scenarios.

4.3 Choices of the Regularization Parameter

As discussed in the preceding subsections, Tikhonov regularization ensures that the regularized solution converges to the true solution asymptotically as $\delta \rightarrow 0$. However, in practical applications, the ideal limit of vanishing noise ($\delta \rightarrow 0$) is rarely attainable. Consequently, we must determine an appropriate value of α for a finite noise level δ . This necessitates a balance: α must be chosen neither too large nor too small. If α is too small, it fails to provide adequate regularization, and the solution retains the ill-posed characteristics of the original problem. For instance, the solution may become unstable and highly oscillatory. Conversely, if α is too large, the approximation error increases significantly. In this case, the penalty term dominates the Tikhonov functional, leading to an oversmoothed solution that fails to capture the primary features of the input data. Hence, a rigorous and systematic selection of α is essential.

Formally, strategies for selecting the regularization parameter α are generally classified into two broad categories: a priori and a posteriori methods. In the a priori approach, α is determined prior to minimizing the Tikhonov functional, relying on the knowledge of the true solution. This approach offers a theoretical guarantee of convergence. For instance, under the source condition $f = K^* K z$ with $\|z\|_F \leq E$, setting $\alpha = (\delta/E)^{2/3}$ yields the error estimate $\|f_\alpha^\delta - f\| \leq \frac{3}{2} \delta^{2/3} E^{1/2}$, which converges to zero as $\delta \rightarrow 0$. This result relies on two a priori assumptions: the boundedness of $\|z\|_F$ by E , and the smoothness of the true solution f inherited from the smoothing operators K and K^* . Despite its theoretical soundness, the a priori strategy is often impractical in real physical applications. This is because essential constants, such as E , are rarely known. This limitation motivates the use of a posteriori methods, which determine α

based solely on the noisy data g^δ and/or an estimate of the noise level δ .

In the a posteriori approach, the regularization parameter α is varied over a wide range of values. For each candidate value α_i , a regularized solution $f_{\alpha_i}^\delta$ is obtained via Tikhonov regularization, thereby generating a family of solutions $\{f_{\alpha_i}^\delta\}$. Subsequently, a specific selection criterion is applied to identify the optimal parameter α and its corresponding solution f_α^δ from this set. Widely used and effective criteria include the Discrepancy Principle, the Quasi-Optimality Criterion, and so on.

Before introducing specific criteria, it is important to note that we expect the physical solution to be robust and not overly sensitive to the specific value of α . Ideally, there should exist a “plateau” in the parameter space where the solution remains stable. We adopt a belt and braces approach to select α and the corresponding solution. Specifically, the choice of α is determined by the rigorous mathematical principles introduced below, while simultaneously requiring the existence of a stability plateau. This dual requirement ensures robustness, making the solution more reliable. This expectation will be demonstrated through the toy model analyses presented in the subsequent section. At this stage, if such a plateau is not observed, we must either modify the regularization method or revise the selection criteria. Furthermore, we expect the regularization method and the selection criteria to remain consistent within the same class of physical problems. However, it may be necessary to adapt the selection criteria or even change the regularization method entirely, if the nature of the problem changes, for instance, transitioning from continuous to discrete problems, or from single-peak to multi-peak structures.

(1) Discrepancy Principle

Regularization methods essentially seek a balance between data fidelity and solution stability. Given a noise level δ , the residual $\|Kf_\alpha^\delta - g^\delta\|$ should be neither excessively large nor small. If $\|Kf_\alpha^\delta - g^\delta\| \ll \delta$, the model fits the observed data more closely than the data’s own precision allows. This indicates an overfitting of the noise, which implies that the regularization parameter is too small. Conversely, if $\|Kf_\alpha^\delta - g^\delta\| \gg \delta$, the fit of the data is poor, failing to capture the dominant features of the signal. This represents an underfit, indicating that the regularization parameter is too large. The discrepancy principle aims to find a value of α such that $\|Kf_\alpha^\delta - g^\delta\| \approx \delta$. This ensures an appropriate balance between suppressing noise amplification and retaining effective information.

The discrepancy principle is conceptually analogous to the criterion of a χ^2 fit. In statistical estimation, a model is generally considered well-fitted when the reduced chi-squared statistic, $\chi_{\min}^2/\text{d.o.f.}$, is approximately unity. A value significantly smaller than one suggests overfitting, whereas a value substantially larger than one indicates a poor fit. Consequently, the optimal fit corresponds to $\chi_{\min}^2/\text{d.o.f.} \approx 1$. In the context of Tikhonov regularization, this principle motivates the selection of the regularization parameter α such that the residual norm matches the noise level, i.e., $\|Kf_\alpha^\delta - g^\delta\| = \delta$.

Theorem 4.4 (Discrepancy Principle). *Let $K : F \rightarrow G$ be linear, compact and one-to-one with dense range in G . Let $Kf = g$ with $f \in F, g \in G, g^\delta \in G$ such that $\|g - g^\delta\| \leq \delta < \|g^\delta\|$. Let the Tikhonov solution f_α^δ satisfy*

$$\|Kf_\alpha^\delta - g^\delta\| = \delta. \quad (4.17)$$

Then $f_\alpha^\delta \rightarrow f$ for $\delta \rightarrow 0$. That is, the discrepancy principle is admissible.

Its proof is provided in Appendix B.

The most significant advantage of the discrepancy principle lies in its rigorous theoretical foundation; specifically, it possesses a strict proof of convergence, guaranteeing that the regularized solution converges to the true solution as the noise level δ approaches zero. In contrast, establishing convergence for a posteriori methods is often a formidable challenge. As will be demonstrated subsequently, widely used techniques such as the quasi-optimality criterion and the L-curve criterion currently lack such proofs of convergence. Consequently, the discrepancy principle is prioritized in this work due to its superior mathematical reliability.

On the other hand, the limitation of the discrepancy principle is evident. The criterion $\|Kf_\alpha^\delta - g^\delta\| = \delta$ requires precise knowledge of the noise level δ , a condition that is rarely met in practice. Since δ is typically unknown, any variation in its estimation directly affects the regularization parameter α , thereby altering the solution. Consequently, the discrepancy principle is primarily applicable when δ is known with certainty, such as in experimental data or lattice QCD where rigorous error analysis is available.

(2) Quasi-Optimality Criterion

When the noise error δ is unknown, the quasi-optimality criterion can be employed to determine the value of α via:

$$\alpha_{\text{opt}} = \arg \min_{\alpha > 0} \left\{ \left\| \alpha \frac{df_\alpha^\delta}{d\alpha} \right\| \right\}. \quad (4.18)$$

The underlying rationale is as follows. First, one seeks a plateau of α by minimizing $df_\alpha^\delta/d\alpha$. Second, preference is given to smaller values of α , which correspond to a smaller residual (i.e., a smaller regularization approximation error). The quasi-optimality criterion minimizes the difference between adjacent regularized solutions. This approach ensures the robustness and insensitivity of the physical results to the specific value of α . Consequently, it is widely used in practical applications.

The quasi-optimality criterion offers the distinct advantage of not requiring an exact value for the noise level δ . When the solution landscape exhibits a single plateau with respect to α , the criterion effectively identifies it. In cases with multiple plateaus, the method inherently favors the one corresponding to a smaller α . This approach aims to achieve higher resolution, representing a strategy that prioritizes fidelity. However, the success of this strategy depends on the requirement that the noise level be sufficiently low.

When input errors are significant, complications arise. If the quasi-optimality criterion selects an excessively small α , the constraining power of the penalty term becomes too weak. Consequently, the regularization process is dominated almost entirely by the data fidelity term. Under such conditions, a model with such a weak restricting effect fails to distinguish between the true signal and random noise. The resulting solution often exhibits spurious high-frequency oscillations induced by the noise, a phenomenon known as under-regularization. Therefore, in practice, this criterion often requires a combination with other methods or manual judgment to ensure robust results.

4.4 Discretization for Tikhonov Regularization

The Tikhonov regularization method seeks the minimizer of the functional defined in Eq. (4.16). However, obtaining an analytical expression for this minimizer is generally infeasible, as the functional is defined over the continuous space F in Eq. (2.6), such as $F = L^2(t_{\min}, \Lambda)$ or $H^1(t_{\min}, \Lambda)$. Therefore, it is necessary to discretize the functional to compute a numerical solution. This can be achieved by constructing a finite-dimensional subspace $X_n = \text{span}\{\varphi_0, \varphi_1, \dots, \varphi_n\}$ of F and solving the problem within this subspace. Finally, we demonstrate that the solution in the discretized space converges to the true solution as the space dimension approaches infinity.

To discretize the Tikhonov functional, a finite element method based on piecewise linear basis functions is employed. The interval $[t_{\min}, \Lambda]$ is divided into uniform subintervals n by grid points defined as

$$x_i = t_{\min} + ih, \quad i = 0, 1, 2, \dots, n, \quad (4.19)$$

where the step size is $h = (\Lambda - t_{\min})/n$. On this grid, the piecewise linear hat functions are defined in the following

$$\varphi_i(x) = \begin{cases} \frac{x-x_{i-1}}{h}, & x \in [x_{i-1}, x_i], \\ -\frac{x-x_{i+1}}{h}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise}, \end{cases} \quad \varphi_0(x) = \begin{cases} -\frac{x-x_1}{h}, & x \in [x_0, x_1], \\ 0, & \text{otherwise}, \end{cases} \quad \varphi_n(x) = \begin{cases} \frac{x-x_{n-1}}{h}, & x \in [x_{n-1}, x_n], \\ 0, & \text{otherwise}, \end{cases} \quad (4.20)$$

for $i = 1, \dots, n-1$. These functions form a basis of the subspace X_n . The continuous minimization problem is then restricted to X_n , leading to the discrete problem

$$f_{\alpha,n}^\delta = \arg \min_{f \in X_n} J_\alpha(f) = \|Kf - g^\delta\|_G^2 + \alpha \|f\|_F^2. \quad (4.21)$$

The approximate solution is then expressed as

$$f_{\alpha,n}^\delta(x) = \sum_{i=0}^n c_i \varphi_i(x), \quad (4.22)$$

with the coefficients c_i to be determined. Substituting it into the functional $J_\alpha(f)$ leads to

$$\begin{aligned} J_\alpha(f_{\alpha,n}^\delta) &= \left\| \sum_{i=0}^n c_i K\varphi_i - g^\delta \right\|_G^2 + \alpha \left\| \sum_{i=0}^n c_i \varphi_i \right\|_F^2 \\ &= \sum_{i,j=0}^n c_i c_j (K\varphi_i, K\varphi_j)_G - 2 \sum_{i=0}^n c_i (K\varphi_i, g^\delta)_G + (g^\delta, g^\delta)_G + \alpha \sum_{i,j=0}^n c_i c_j (\varphi_i, \varphi_j)_F. \end{aligned} \quad (4.23)$$

In order to find the extreme values of this multivariate function, we set the partial derivative of $J_\alpha(f_{\alpha,n}^\delta)$ with respect to each coefficient c_i ($i = 0, 1, \dots, n$) equal to zero. This results in the following linear system:

$$(A + \alpha B)C = D, \quad (4.24)$$

where

$$A_{ij} = (K\varphi_i, K\varphi_j)_G, \quad B_{ij} = (\varphi_i, \varphi_j)_F, \quad C_j = (c_0, c_1, \dots, c_n)^T, \quad D_i = (K\varphi_i, g^\delta)_G^T. \quad (4.25)$$

The coefficient vector C is obtained by solving the linear system, which gives the numerical minimizer of the Tikhonov functional $f_{\alpha,n}^\delta(x)$ in Eq. (4.22). Note that, due to the continuity of the finite element basis functions in Eq. (4.20), the inner products in Eq. (4.25) are still defined in the continuous spaces F and G .

Alternatively, the above linear system is equivalently obtained by projecting the regularized normal equation, $(K^*K + \alpha I)f_\alpha^\delta = K^*g^\delta$, onto the basis functions φ_i via the inner product,

$$\left(\varphi_i, (K^*K + \alpha I) \sum_j c_j \varphi_j \right)_F = (\varphi_i, K^*g^\delta)_F. \quad (4.26)$$

Then we obtain

$$\sum_j c_j (K\varphi_i, K\varphi_j)_G + \sum_j c_j \alpha (\varphi_i, \varphi_j)_F = (K\varphi_i, g^\delta)_G, \quad (4.27)$$

which is equivalent to Eq. (4.24).

While discretization inherently introduces approximation errors, the convergence and reliability of this numerical solution are guaranteed by Theorem B. Specifically, as $n \rightarrow \infty$, the numerical solution $f_{\alpha,n}^\delta$ converges to the theoretical solution f_α^δ , ensuring that numerical errors can be controlled through mesh refinement.

Theorem 4.5. *For fixed noise level δ and regularization parameter $\alpha > 0$, we have*

$$\|f_{\alpha,n}^\delta - f_\alpha^\delta\|_F \rightarrow 0, \quad n \rightarrow \infty. \quad (4.28)$$

The proof of this theorem is given in Appendix B.

In summary, this section has provided a comprehensive treatment of the Tikhonov regularization method and the associated strategies for determining the value of the regularization parameter α . A

rigorous theoretical foundation has been established, confirming that the Tikhonov approach effectively resolves the ill-posedness inherent in the inverse problem of dispersion relation. It guarantees a stable and unique regularized solution that converges to the true solution as the input error tends to zero. Importantly, the overall solution process is built upon stringent mathematical principles and introduces no artificial or model-dependent assumptions.

From the error bound given in Eq. (4.5), $\|f_\alpha^\delta - f\| \leq \|R_\alpha K f - f\| + \delta \|R_\alpha\|$, it is straightforward to analyze the uncertainties in the solution. The bound consists of two parts: the error from the input data amplified by $\|R_\alpha\|$, and the error due to the regularization approximation. The convergence result $\|f_\alpha^\delta - f\| \rightarrow 0$ and $\delta \rightarrow 0$ indicates that the solution can be systematically improved by reducing the input error δ . We will present the numerical results and the corresponding error analysis in the next section, using three toy models.

5 Tests of Toy Models

In the previous section, the mathematical framework of the inverse problem approach is introduced. Before tackling detailed physical problems, some toy models can be tested to intuitively examine the effect of solving the inverse problem for the dispersion relation.

5.1 Three Toy Models

To evaluate the effectiveness of the Tikhonov regularization method, we employ three representative toy models:

$$\begin{aligned}
 \textbf{Model 1 (monotonic)} : f(x) &= a_1 2 \log(x + 0.5) + \frac{a_2}{x + 3}, \\
 \textbf{Model 2 (simple non-monotonic)} : f(x) &= a_1 4 x e^{-a_2 x}, \\
 \textbf{Model 3 (resonance)} : f(x) &= \frac{1}{\pi} \frac{a_1 m \Gamma}{(x - m^2)^2 + (a_1 m \Gamma)^2} + a_2 \frac{x}{5}.
 \end{aligned} \tag{5.1}$$

These models are designed to serve as benchmarks for a broad class of physical problems, with $a_1 = a_2 = 1.0$. For instance, Models 1 and 2 are analogous to continuum spectral functions, which often exhibit simple monotonic or non-monotonic behaviors, as well as parton distribution functions and distribution amplitudes. Model 2 converges to zero as x tends to infinite, while model 1 does not. Besides, models 1 and 2 capture the typical behaviors of exponential and logarithmic functions. Model 3 represents resonance phenomena, with $m = 0.8$ and $\Gamma = 0.5$. The primary objective is to assess the performance of the regularization method, especially in scenarios where the input data are subject to errors represented by the parameters $a_{1,2}$.

The numerical procedure is illustrated in Fig. 5. For each toy model, the true solution $f(x)$ is

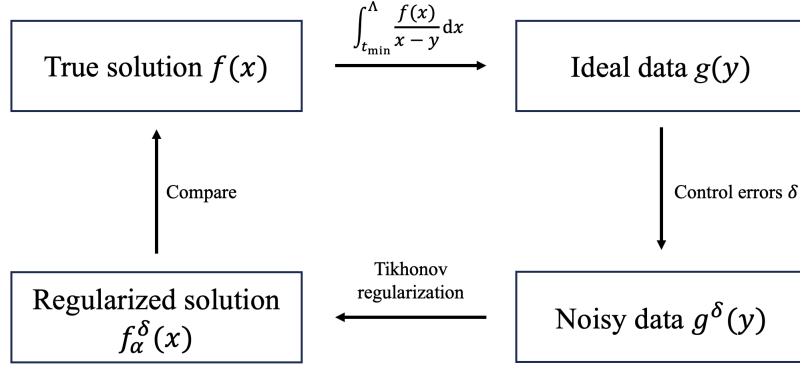


Figure 5: Numerical procedure for evaluating the effectiveness of the inverse problem approach.

transformed by the dispersion integral to obtain the exact $g(y)$. Controlled errors are then added to simulate real physics, resulting in noisy data $g^\delta(y)$. Tikhonov regularization is applied to invert $g^\delta(y)$, producing the regularized solution $f_\alpha^\delta(x)$, which is then compared to the original $f(x)$ to assess the effectiveness of the method.

Two types of errors in $g^\delta(y)$ are considered. The first approach adds random noise to each discrete point of $g(y_i)$, which closely mimics the experimental data. However, in the present inverse problem approach, the input data is always obtained from perturbative calculations. Therefore, a second approach is adopted, where errors enter through analytic functions with uncertain input parameters (e.g., the energy scale or vacuum condensates). The resulting error formulas are expressed analytically and this type of input error is tested.

Specifically, the input data are given by $g^\delta(y, a_1, a_2) = \int_{t_{\min}}^{\Lambda} \frac{f(x, a_1, a_2)}{x-y} dx$, where $a_{1,2}$ are explicitly expressed from $f(x)$ in Eq. (5.1). The values of $a_{1,2}$ are randomly sampled from Gaussian distributions with mean μ_i and standard deviation σ_i ($i = 1, 2$). Here, $\mu_{1,2} = 1.0$, and $\sigma_{1,2}$ take the values 0.3, 0.1, and 0.01, corresponding to the input error levels of 30%, 10%, and 1%, respectively. The parameters in Eq. (2.5) are taken as $t_{\min} = 0$, $\Lambda = 2$, $q_{\text{low}}^2 = -100$, $q_{\text{up}}^2 = -5$, with the discretization step sizes $h_x = 0.005$ in Eq. (4.19) and $h_y = 0.01$.

5.2 Numerical Results Under Regularization

The efficacy of the regularization schemes is validated using toy models. The boundary conditions are introduced prior to the detailed methodology. Since the original integral equation in Eq. (2.4) separates perturbative and non-perturbative contributions, the continuity of the imaginary part of the correlation function serves as a boundary condition for the inverse problem. More prior knowledge leads to better inversion results. Therefore, the boundary conditions on $f(x)$ should be exploited to improve the inversion in all toy models. Let the boundary conditions be $f(t_{\min}) = M$ and $f(\Lambda) = N$. The function $f(x)$ can then

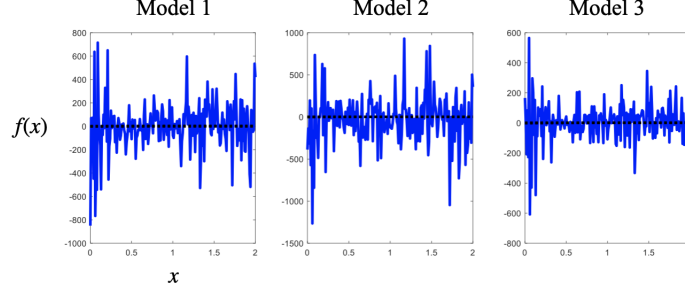


Figure 6: Solutions without regularization (blue lines) compared to the exact solution (black lines).

be decomposed as $f(x) = u(x) + v(x)$, where $v(x)$ is linear and satisfies the same boundary conditions, i.e., $v(x) = M \frac{(x-\Lambda)}{t_{\min}-\Lambda} + N \frac{(x-t_{\min})}{\Lambda-t_{\min}}$. The unknown function $u(x)$ is then solved from the modified integral equation:

$$\int_{t_{\min}}^{\Lambda} \frac{u(x)}{x-y} dx = g^{\delta} - \int_{t_{\min}}^{\Lambda} \frac{v(x)}{x-y} dx := G^{\delta}$$

with homogeneous boundary conditions $u(t_{\min}) = u(\Lambda) = 0$. Applying a regularization method yields the regularized solution u_{α}^{δ} , and the final solution is reconstructed as $f_{\alpha}^{\delta}(x) = u_{\alpha}^{\delta}(x) + v(x)$.

Without Regularization

First, the necessity of regularization for the ill-posed inverse problem is demonstrated by no regularization. To this end, the penalty term is neglected in the Tikhonov functional in Eq. (4.15), retaining only the residual term, equivalent to the conventional χ^2 fitting approach. In minimizing the functional, $K f_{\alpha}^{\delta}$ becomes infinitely close to g^{δ} .

The results without regularization are shown in Fig. 6 for the three toy models. The numerical results (blue lines) exhibit high-frequency oscillations and are out of control compared to the true values (black lines). This instability clearly illustrates the ill-posedness of the original problems. Without regularization, physically meaningful solutions cannot be obtained for such ill-posed problems.

Results under Tikhonov Regularization in the L^2 space

The numerical results obtained from Tikhonov regularization are presented using Eq. (4.16) with the solution space F taken as the L^2 space. The regularization parameter α is scanned from 1×10^{-1} to 1×10^{-19} to examine its influence on the solution, and the input error is set to a 10% deviation ($a_1 = a_2 = 1.1$). The results are shown in Fig. 7.

Analyzing these results, regularization is found to yield considerably improved solutions compared to the highly oscillatory ones shown in Fig. 6. With an appropriate choice of α , the solution approaches the true value, confirming both the necessity and effectiveness of regularization for ill-posed problems.

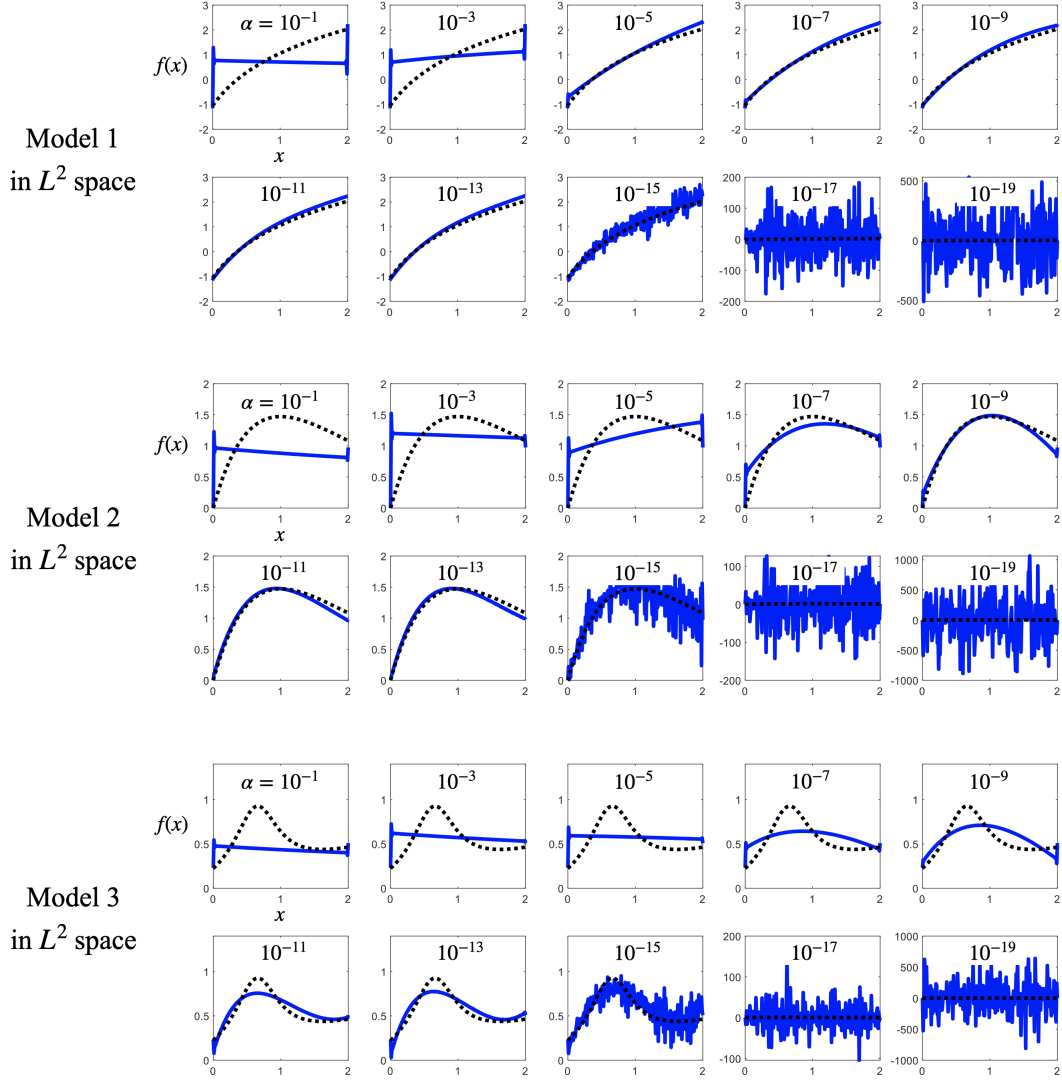


Figure 7: Numerical solutions for the three toy models using Tikhonov regularization with the solution space $F = L^2(t_{\min}, \Lambda)$. Each figure displays the regularized solutions $f_{\alpha}^{\delta}(x)$ (blue solid) and the exact solutions $f(x)$ (black dashed), with the corresponding α value at the top.

It is also observed that α must be chosen within an appropriate range. A value of α that is too large leads to poor approximation and significant deviations, while a value that is too small renders regularization ineffective, gradually restoring ill-posedness and causing oscillations and instability. Satisfactory results are achieved for intermediate α values, which is fully consistent with the expectations discussed in the preceding section.

Notably, within this intermediate regime, the results are not particularly sensitive to α . Values spanning several orders of magnitude yield comparable outcomes. This is a highly favorable feature from a physical perspective and will be discussed in detail in the following section.

Approximate solutions are close to the true solution for two fundamental reasons. First, the true solution resides in a constrained subspace, not randomly distributed across the entire solution space. Second, physical signals are inherently finite, while unphysical and noise-induced oscillations tend to cause the solution norm to diverge. Thus, least-norm solutions naturally favor the true physical state.

This explains why highly oscillatory solutions induced by small input errors are excluded from the solution set. Such oscillations are not part of the true solution. Rather, they are mathematical artifacts arising from the fact that a smooth operator has an unbounded inverse, which amplifies small data errors into large high-frequency oscillations. Regularization removes these artifacts by incorporating prior knowledge. The Low-frequency components are dominated by true signals and are retained, whereas the high-frequency components are dominated by noise and are discarded. Since infinitely high-frequency oscillations do not exist in most real physical systems, they are essentially hallucinations generated by finite errors acting on ill-posed operators. Dispelling these illusions reveals the truth.

Results under Tikhonov Regularization in the H^1 space

From Fig. 7, the solutions in the end-point region exhibit oscillations for some large values of α in each model. This is due to the use of boundary conditions and the fact that the L^2 space requires only square integrability, not smoothness. In many physical problems, solution functions are smooth, analogous to the three toy models. Therefore, we can enforce a prior smoothness condition by changing the solution space from L^2 to H^1 (see Appendix A for definitions). In the H^1 space, the solution function is first-order differentiable, which will improve the solution to our problem.

The numerical results in the H^1 space are presented in Fig. 8. Compared to the L^2 space results in Fig. 7, all three models show improvement, exhibiting no oscillations in the end-point regions. Models 1 and 2 display a wider plateau with respect to α . For Model 3, improved solutions are obtained at $\alpha = 10^{-13}$ and 10^{-15} , which are consistent with the true solution given an uncertainty of 10%.

As smoothness is a common a priori condition in physical applications, we adopt the H^1 space in the subsequent discussion, though the L^2 space would be used in practical cases where smoothness is not

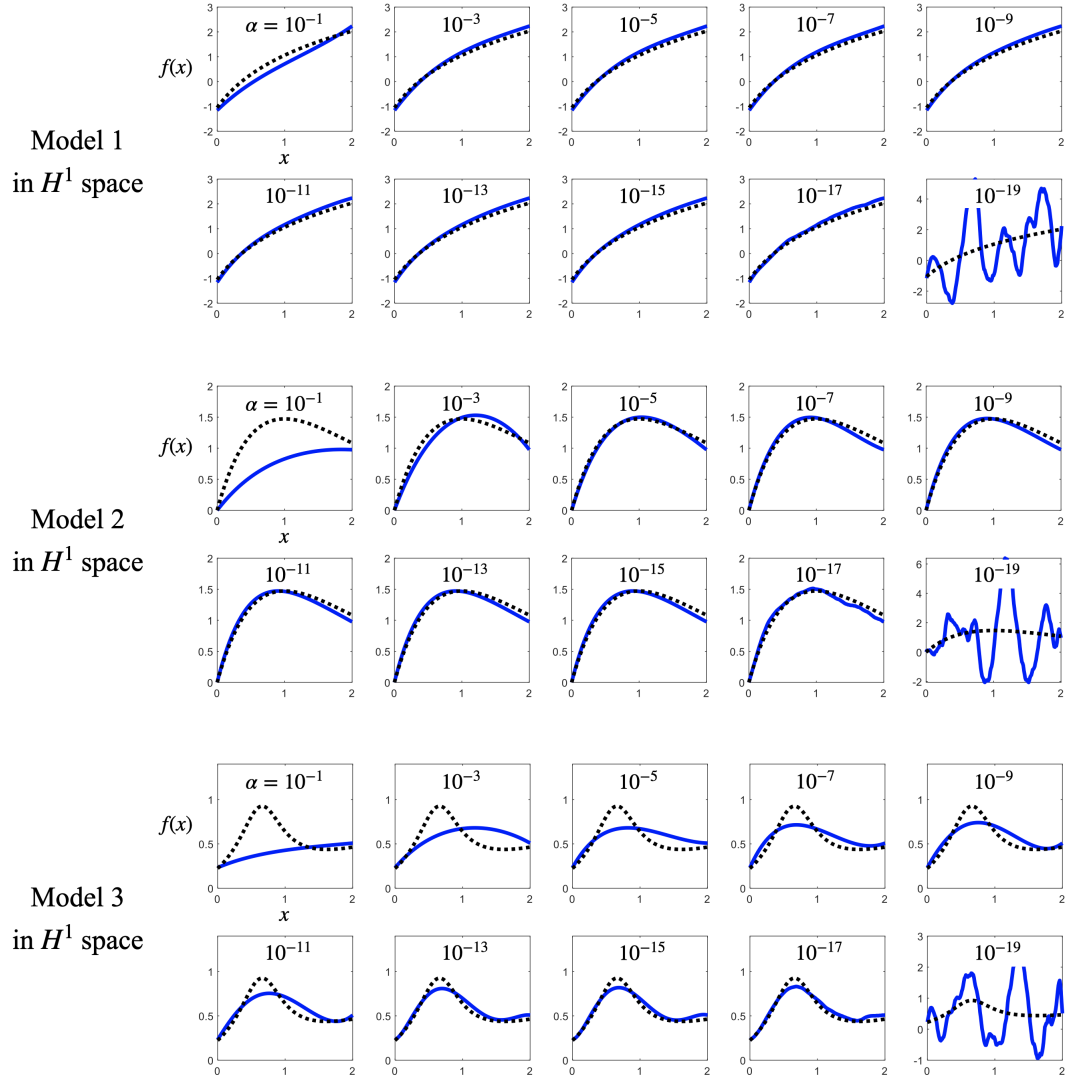


Figure 8: Same as Fig. 7 but for the solution space $F = H^1(t_{\min}, \Lambda)$.

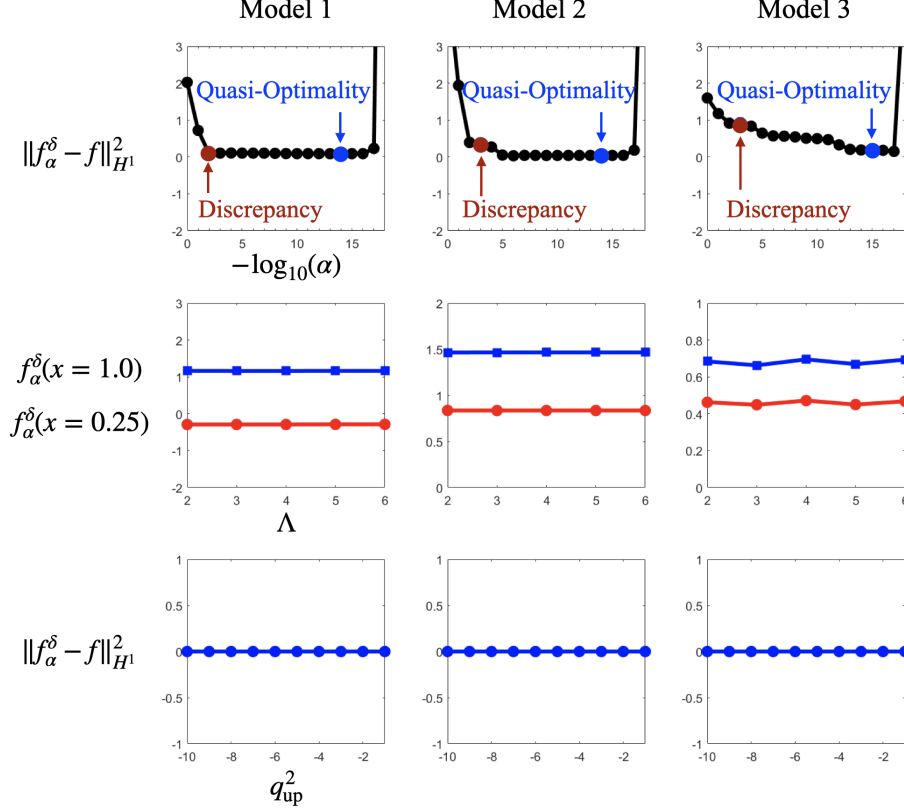


Figure 9: Stability plateaus of the regularization parameter α (top), the perturbative-nonperturbative boundary Λ (middle), and the upper bound of the input range q_{up}^2 (bottom) are shown for the three toy models (left to right).

satisfied.

Stability Plateaus of Parameters

The sensitivity of the results to all parameters in the inverse problem approach is examined here. In the regularization method, the primary parameter is α . As observed in Fig. 8, the results exhibit a broad stability plateau with respect to α . This behavior is investigated in detail below. Additionally, Eq. (2.5) presents the integral equation for the inverse problem of the dispersion relation. This equation involves the scale parameter Λ , which demarcates the perturbative and non-perturbative regimes, as well as the integration limits q_{up}^2 and q_{low}^2 . The dependence of the results on these parameters is also analyzed. The numerical results of the tests are shown in Fig. 9.

We first examine the dependence of the results on α , using the norm of the difference between the regularized solution and the exact solution, $\|f_\alpha^\delta - f\|_{H^1}^2$. Clear plateaus spanning several orders of magnitude is observed, as shown at the top of Fig. 9. Models 1 and 2 exhibit relatively wide plateaus, ranging from $\alpha = 10^{-2}$ to 10^{-16} for model 1 and from 10^{-5} to 10^{-16} for model 2, while model 3 shows a narrow plateau from 10^{-13} to 10^{-17} . These observations are consistent with the mathematical properties

and meet our physical expectations. Although α cannot be extremely large or extremely small, a very wide range of intermediate values is acceptable, ensuring that the results remain insensitive to α . For simple problems such as model 1, different selection criteria choose values of α within the same plateau, demonstrating that the results are highly robust to the choice of α .

Furthermore, the values of α determined by the quasi-optimality criterion are always significantly lower than those obtained from the discrepancy principle. The former typically yields the best solutions for all three models. Although the discrepancy principle possesses the favorable mathematical property of proven convergence, it is generally more appropriate for scenarios with minimal input errors (e.g., smaller than 0.1%), as discussed in the next subsection. Given the substantial input errors (1% – 30%), the quasi-optimality criterion is preferred to determine α , and will be used in the following discussions unless otherwise specified.

The dependence of the results on Λ is shown in the middle of Fig. 9, using the solutions at $x = 0.25$ and $x = 1.0$ as examples. Plateaus are observed in $\Lambda \in [2, 6]$, which is consistent with our expectations, since the transition from perturbative to nonperturbative regimes is continuous and smooth. In practical physical problems, the plateau of Λ might not be as wide as those in toy models. A Λ that is too large increases the amount of unknown information to be solved, thereby complicating the problem. In contrast, a Λ that is too small introduces larger nonperturbative errors in the input term of Eq. (2.4), leading to deteriorated results.

The influence of the input range is shown at the bottom of Fig. 9, where $q_{\text{up}}^2 \in [-10, -1]$ with $q_{\text{low}}^2 = -100$. The resulting plateaus are excellent. Although not shown, fixing q_{up}^2 and varying q_{low}^2 also yields excellent plateaus. As long as the input information is sufficiently good, the results remain insensitive to the choice of input range.

Finally, we discuss model 3. The plateau of α in model 3 is narrower than those in models 1 and 2. Furthermore, the dependence on Λ in model 3 exhibits slight fluctuations, in contrast to the very flat behavior observed in models 1 and 2. We have also examined that reducing the width of the Breit-Wigner peak in model 3 leads to further deterioration of the results. In the case of two or more peaks, solving the problem becomes even impossible.

These observations indicate that fine structures, such as Breit-Wigner peaks, are inherently more difficult to resolve under Tikhonov regularization. Although other regularization methods may better handle such cases, a detailed investigation lies beyond the scope of this paper, and therefore we continue to focus on Tikhonov regularization. More importantly, this finding suggests that no single regularization method is universally applicable. Under a given scheme, certain problems cannot be solved satisfactorily. In other words, regularization methods are not a panacea. Without further refinements, some problems

may remain inherently unsolvable.

5.3 Ability to Analyze Uncertainties and Systematically Improve Precision

Rigorous uncertainty analysis is essential for theory-experiment comparisons and the search for new physics. The ability to systematically improve precision is also a prerequisite for first-principles calculations. For instance, perturbative QCD can achieve higher precision through higher-order corrections, while lattice QCD can do so by increasing computational resources. In this subsection, we demonstrate that the inverse problem approach enables the rigorous analysis of both statistical and systematic uncertainties, as well as the controllable improvement of precision. This improvement is reflected in two aspects: (1) reducing the input error δ , and (2) improving the prior information and regularization methods, as illustrated in Fig. 3.

Uncertainty Analysis

From Eq. (4.5), the uncertainty of solution consists of two parts:

$$\|f_\alpha^\delta - f\| \leq \delta \|R_\alpha\| + \|R_\alpha K f - f\|.$$

The first term $\delta \|R_\alpha\|$ represents the statistical uncertainty arising from the input error $\|g^\delta - g\| \leq \delta$. To analyze this statistical uncertainty in detail, α is first fixed by solving the inverse problem for the central curve of $g^\delta(y)$. Then, all possible values of the input parameters (e.g., a_1, a_2) within the error bound are considered. For each of such g_i^δ , the corresponding solution $f_{\alpha,i}^\delta(x)$ is computed. Repeating this process for all different inputs g_i^δ yields an ensemble of solutions, from which the statistical mean and the associated standard deviation are computed at each point x .

The second term $\|R_\alpha K f - f\|$ is the systematic uncertainty from the regularization approximation. It is the difference between the regularized solution $R_\alpha K f$ and the exact solution f . Since the exact solution f is typically unknown, it can be replaced in practice by the regularized solution f_α^δ . Note that δ and α are fixed here. We then examine the difference between $R_\alpha K f_\alpha^\delta$ and f_α^δ .

A key advantage of the inverse problem approach is its ability to rigorously quantify systematic uncertainty. Yet model-dependent methods typically cannot do this precisely. Theoretically, an admissible regularization requires $\alpha(\delta) \rightarrow 0$ as $\delta \rightarrow 0$ (see Definition 4.2). The discrepancy principle makes this relationship explicit: α decreases with δ . Hence, a smaller δ leads to a smaller α and consequently to a smaller systematic uncertainty. This chain of reasoning shows how the inverse problem approach can systematically improve precision, as illustrated in the following.

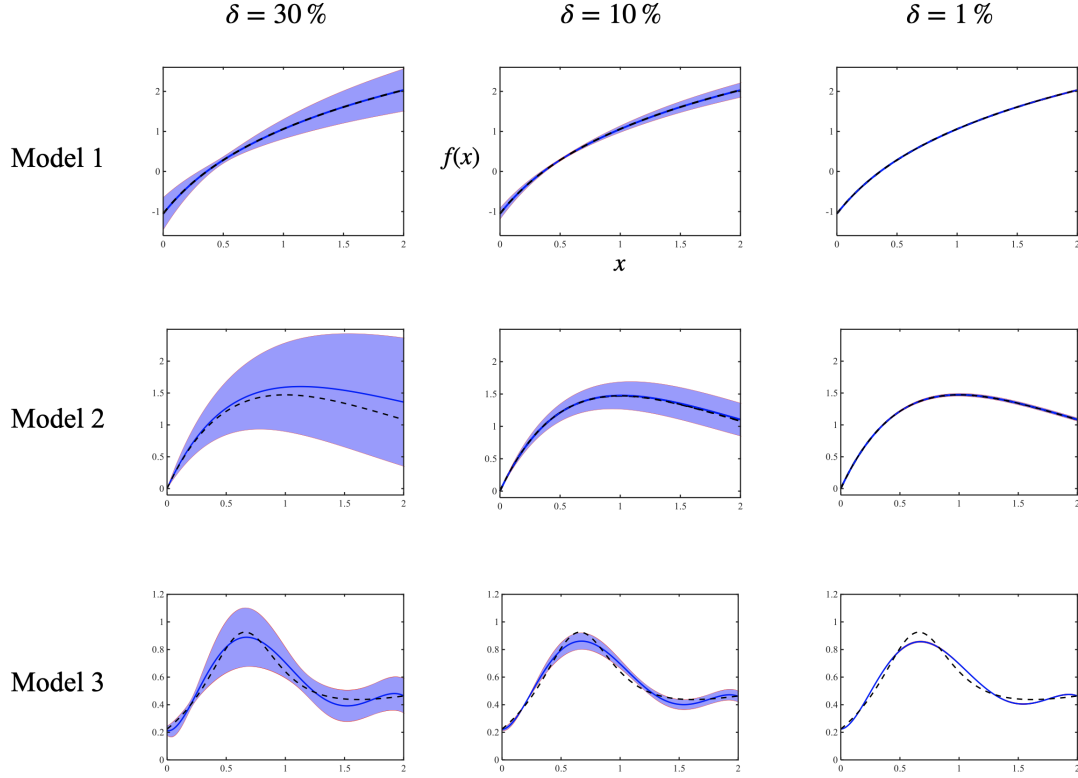


Figure 10: Numerical results under three levels of input errors. The central values f_{α}^{δ} (blue solid curve) with statistical uncertainties (purple band) and systematic uncertainties (red band, lying outside the purple band) are compared against the true solutions (black dashed curve). Rows: models 1, 2, 3 (top to bottom). Columns: input errors 30%, 10%, 1% (left to right).

Improvement of Precision (I): Reducing Input Errors

Having established a reasonable method for uncertainty analysis, we now examine how the solution behaves under different levels of input error. Three levels of input error (30%, 10%, and 1%) are examined. The numerical results are shown in Fig. 10. The central values f_{α}^{δ} represented by solid blue curves, together with their statistical uncertainties indicated by purple bands and systematic uncertainties shown as red bands outside the purple bands, are compared against the true solutions represented by black dashed curves.

The statistical uncertainties of the solutions are approximately 30%, 10%, and 1%, respectively, corresponding to the three levels of input errors. It is evident that the statistical uncertainties diminish with decreasing input errors, as expected. This concretely demonstrates that the solution precision improves as the input errors decrease. In addition, a clear convergence of the solutions is observed in models 1 and 2. In particular, for the input error of 1%, the approximate solutions converge almost entirely to the true solutions.

For model 3 with large input errors, the true solution lies within the uncertainty region. In contrast, when the input error is reduced to 1%, the uncertainty narrows but exhibits a noticeable deviation from the true solution. This once again demonstrates the limited power of Tikhonov regularization in addressing fine-structure problems like model 3. In this case, alternative regularization methods are required.

The systematic uncertainties are too small to be clearly visible in the figures, because α is determined to be 10^{-14} or 10^{-15} by the quasi-optimality criterion. The numerical results show that the systematic uncertainties are of the order of 10^{-3} or 10^{-4} , which is much lower than the statistical uncertainties of 30%–1%. Consequently, the overall uncertainties are dominated by the statistical ones.

In addition, the values of α determined by the quasi-optimality criterion are almost unchanged when the input errors vary, i.e., they do not satisfy convergence. This is not a problem for input errors as large as 30% – 1%, where the systematic uncertainties are much smaller than the statistical ones. When input errors are further reduced, e.g., to 10^{-3} – 10^{-5} , the discrepancy principle applies. However, since such low error levels lack a clear physical correspondence, we do not explicitly show the numerical results in this work. The values of α from the discrepancy principle decrease with the input errors, leading to a reduction in systematic uncertainties. Consequently, both statistical and systematic uncertainties diminish as input errors decrease, indicating convergence of solutions, as proved in Theorem 4.4. Thus, the precision of solutions in the inverse problem approach can be systematically improved by reducing input errors.

Improvement of Precision (II): Improving Prior Information and Regularization Method

From the perspective of minimizing a regularized functional, solving an ill-posed inverse problem involves a trade-off between two competing requirements: the penalty term encodes prior knowledge that controls the desired properties of the solution, whereas the residual term ensures that the solution remains faithful to the observed data. The regularized functional can be defined as [23]

$$J_\alpha(f) = \|Kf - g^\delta\|_G^2 + \alpha \Omega[f], \quad (5.2)$$

where $\Omega[f]$ is an appropriate norm that encodes prior information about the solution. For example, $\Omega[f] = \|f\|_F^2$ in Tikhonov regularization, where the norm is defined on the space F to characterize the properties of the solution f and to control its stability. Refining the prior information allows us to improve the definition of the functional in the regularization method, consequently leading to a more accurate solution.

The most intuitive improvement is the transition from the L^2 space to the H^1 space discussed in Sec. 5.2. When $F = L^2(t_{\min}, \Lambda)$, only the boundedness of the solution is imposed, without any smoothness requirement. The resulting solutions are shown in Fig. 7, where oscillations appear at the left and right

endpoints, or over the entire domain, for certain values of α . When the smoothness prior is added, i.e., $F = H^1(t_{\min}, \Lambda)$, the solutions are significantly improved, as shown in Fig. 8. In this case, the oscillations disappear and the plateau of α becomes substantially wider.

As a counterexample, Figure 10 shows that for Model 3 at 1% input error, the obtained solution does not fully match the true solution. This indicates that Tikhonov regularization is not suitable for problems with fine-structures. The standard Tikhonov regularization, typically formulated in norm spaces L^2 or H^1 , penalizes the energy of the solution or its derivatives integrated over the domain. This effectively suppresses high-frequency components and smooths out the internal fine structures of the solution. Consequently, Tikhonov regularization inherently favors smooth solutions and is less effective for non-smooth problems, where it may fail to recover fine details. The use of alternative regularization methods—such as the L^1 norm commonly used for sparse solutions, iterative regularization methods, or discrete regularization methods—may help solve fine-scale structure problems. However, this lies beyond the scope of this work and is left for future research.

Another type of prior information originates from external constraints, such as precise experimental measurements or accurate calculations from methods like lattice QCD. For instance, when accurate experimental or lattice results are available at a specific energy point within the non-perturbative region, incorporating these constraints into the inverse problem can enhance the accuracy of the solution across the entire region. Similarly, in spectral problems, if the masses and widths of ground-state or excited-state particles have been experimentally measured, such prior information can be leveraged to refine the solution. Therefore, inverse problem methodologies are capable of working in synergy with experiments or lattice QCD to jointly advance the resolution of non-perturbative problems.

Finally, we discuss the influence of prior information on regularization methods and the resulting solution. In general, mathematical techniques alone cannot solve all problems. They cannot compensate for a lack of information, especially when the input data are subject to errors. This is because the information is conserved in the sense that the solution cannot contain more information than is provided by the data and the prior combined. Consequently, having more (and correct) prior information typically improves the quality of the solution. However, highly detailed prior information is not necessary. It often suffices to know qualitative properties of the solution, such as whether it is smooth or contains fine-scale structures like resonances. Then a regularization method that properly encodes such information should be chosen.

6 Conclusion and Outlook

In this work, we have developed a theoretical framework for the inverse problem approach to non-perturbative QCD, supported by rigorous mathematical theorems and proofs. The framework begins with the QFT dispersion relation, separates the high-energy and low-energy scales, and uses known perturbative theories to determine unknown non-perturbative quantities by solving an inverse problem. Generally, we prove that the inverse problem of dispersion relation is ill-posed. It admits a unique but unstable solution. To obtain stable approximate solutions, regularization methods must be utilized. The idea behind regularization is to solve a nearby well-posed problem instead and use its solution as an approximation to the original ill-posed one. We have presented three toy models to vividly illustrate the main features of the inverse problem approach.

We summarize the advantages of the inverse problem approach:

- It is based on rigorous mathematics of regularization theory and does not involve artificial assumptions.
- The regularized solution converges to the true solution as the input error decreases.
- It enables to analyze statistical and systematic uncertainties, and to systematically improve the precision of solution by reducing the input errors and by refining prior information and regularization methods.
- All calculations can be performed on a personal laptop and do not require large computational resources.
- The entire non-perturbative region is calculated simultaneously. Therefore, this approach would benefit the study of excited states and the continuum spectrum.

We point out that inverse problems possess substantial application potential in high-energy physics. Currently, with the continuous improvement in the precision of experimental data, there is an increasing demand for high-precision calculations. Theoretically, high-precision predictions rely on high-order loop and power corrections in perturbative expansions, as well as high-precision inputs for non-perturbative quantities. Inverse problems are expected to serve as an alternative method for determining non-perturbative physical quantities. In the near future, inverse problems are likely to demonstrate their significant power in many areas non-perturbative QCD, such as the parton distributions and the spectroscopy of hadrons.

Further improvements in the dispersion-relation-based inverse problem approach are needed to fully realize these objectives. For example, enhancing the precision of the physical data used as inputs to dispersion integrals is essential. Tikhonov regularization guarantees convergence of the regularized

solution to the true solution as input errors decrease, making precise and reliable data critical. Moreover, the mathematical foundation of inverse problems in high-energy physics requires continued refinement. As an evolving branch of mathematics, inverse problem theory will benefit from more powerful regularization schemes and advanced analytical frameworks, which are essential for probing fundamental solution properties and gaining deeper insights into non-perturbative physics.

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A Basic Functional Analysis

This appendix provides a summary of fundamental definitions and theorems from functional analysis.

Normed spaces (L^2 and H^1 spaces and so on)

The physical problems here are all formulated within the framework of quantum field theory. Therefore, the relevant quantities are naturally defined in a Hilbert space, which is endowed with an inner product structure. In particular, we work within the L^2 and H^1 spaces throughout our analysis, as they provide the appropriate functional framework for describing the physics in its most general formulation.

L^2 space (square-integrable function space) consists of all measurable functions whose squared absolute value has finite integral:

$$\int_a^b |f(x)|^2 dx < \infty. \quad (\text{A.1})$$

H^1 space (Sobolev space) consists of all functions in L^2 whose first weak derivatives also belong to L^2 :

$$\int_a^b |f(x)|^2 + |f'(x)|^2 dx < \infty, \quad (\text{A.2})$$

where $f'(x)$ is the first weak derivative of $f(x)$.

The norms and inner products associated with these spaces are defined as follows. For the $L^2(a, b)$ space:

$$\begin{aligned}\|f\|_{L^2(a,b)} &= \left(\int_a^b |f(x)|^2 dx \right)^{1/2}, \\ (f_1, f_2)_{L^2(a,b)} &= \int_a^b f_1(x)f_2(x) dx.\end{aligned}\tag{A.3}$$

For the $H^1(a, b)$ space:

$$\begin{aligned}\|f\|_{H^1(a,b)} &= \left(\int_a^b (|f(x)|^2 + |f'(x)|^2) dx \right)^{1/2}, \\ (f_1, f_2)_{H^1(a,b)} &= \int_a^b (f_1(x)f_2(x) + f_1'(x)f_2'(x)) dx.\end{aligned}\tag{A.4}$$

In addition, the function space $C[a, b]$ and $L^1[a, b]$ are used in the proof of uniqueness for the inverse problem of dispersion relation.

$C[a, b]$ is the space of all real-valued (or complex-valued) continuous functions defined on the closed interval $[a, b]$. The standard norm on $C[a, b]$ is $\|f\|_{C[a,b]} = \max_{x \in [a,b]} |f(x)|$. An important property of $C[a, b]$ is that for $f_n, f \in C[a, b]$, $\|f_n - f\|_{C[a,b]} \rightarrow 0$ as $n \rightarrow \infty$, is equivalent to f_n converging to f uniformly on $[a, b]$.

$L^1[a, b]$ is the space of all functions whose absolute value has a finite integral, i.e. $\|f\|_{L^1[a,b]} = \int_a^b |f(x)| dx < \infty$.

Linear Bounded and Compact Operators

Definition A.1 (Linear Operator). An operator $K : F \rightarrow G$ is called linear if

$$K(\alpha f_1 + \beta f_2) = \alpha K f_1 + \beta K f_2, \quad \forall f_1, f_2 \in F, \alpha, \beta \in \mathbb{R} \text{ (or } \mathbb{C}).\tag{A.5}$$

Definition A.2 (Bounded Operator). A linear operator $K : F \rightarrow G$ is called bounded if there exists a finite positive number C such that

$$\|Kf\|_G \leq C \|f\|_F,\tag{A.6}$$

for all $f \in F$. Each number C for which this inequality holds is called a bound for the operator K .

Theorem A.3. For a linear operator $K : F \rightarrow G$ mapping a normed space F into normed space G the following properties are equivalent: (a) K is bounded. (b) K is continuous at $f = 0$, i.e., $f_j \rightarrow 0$ implies that $Af_j \rightarrow 0$. (c) K is continuous for every $f \in F$.

Definition A.4 (Operator Norm). Let $K : F \rightarrow G$ be a bounded linear operator. The operator norm of K is defined as the supremum (least upper bound) of $\|Kf\|_G$ over all f in the unit ball:

$$\|K\| := \sup_{\|f\|_F \leq 1} \|Kf\|_G. \quad (\text{A.7})$$

If $f \neq 0$, the operator norm can be equivalently defined as

$$\|K\| := \sup_{f \neq 0} \frac{\|Kf\|_G}{\|f\|_F}. \quad (\text{A.8})$$

This is a normalized perspective. For any non-zero input, we calculate the relative ratio by which the output is amplified, and then take the supremum of this ratio.

Theorem A.5. For a bounded linear operator $K : F \rightarrow G$ and every $f \in F$, and by the definition of the operator norm, the following inequality holds:

$$\|Kf\|_G \leq \|K\| \cdot \|f\|_F. \quad (\text{A.9})$$

Definition A.6 (Adjoint Operator). Let F and G be Hilbert spaces. The adjoint of a bounded linear operator $K : F \rightarrow G$ is the unique operator $K^* : G \rightarrow F$ satisfying the relation

$$(Kf, g)_G = (f, K^*g)_F, \quad (\text{A.10})$$

for all $f \in F$ and $g \in G$.

Definition A.7 (Self-Adjoint Operator). A bounded linear operator $A : F \rightarrow F$ is called self-adjoint if $A = A^*$. Equivalently, it satisfies $(Af, f)_F = (f, Af)_F$ for all $f \in F$.

For a self-adjoint operator A , its norm admits an equivalent characterization via the quadratic form:

$$\|A\| = \sup_{\|f\|=1} |(Af, f)_F|. \quad (\text{A.11})$$

Definition A.8 (Compact Operator). A linear operator $K : F \rightarrow G$ is called compact if it maps each bounded set in F into a relatively compact set in G .

B Proofs of theorems in Sec. 4

In this appendix, we complete the proofs of results in the section of regularization, Sec. 4.

Theorem 4.3. Let $K : F \rightarrow G$ be a bounded linear operator between Hilbert spaces and $\alpha > 0$. Then the Tikhonov functional J_α ,

$$J_\alpha(f) := \|Kf - g\|_G^2 + \alpha \|f\|_F^2 \quad \text{for } f \in F, \quad (\text{B.1})$$

has a unique minimizer $f_\alpha = \arg \min_{f \in F} J_\alpha(f)$. This minimizer f_α is the unique solution of the normal equation

$$\alpha f_\alpha + K^* K f_\alpha = K^* g. \quad (\text{B.2})$$

Proof. Firstly, existence and uniqueness are proven.

Let $\{f_n\} \subset F$ be a minimizing sequence such that $J_\alpha(f_n) \rightarrow J_0 = \inf_{f \in F} J_\alpha(f)$ as $n \rightarrow \infty$. By expanding the squared norms, we have

$$\begin{aligned} J_\alpha(f_n) + J_\alpha(f_m) &= \|Kf_n - g\|^2 + \|Kf_m - g\|^2 + \alpha (\|f_n\|^2 + \|f_m\|^2) \\ &= 2 \left\| K \left(\frac{f_n + f_m}{2} \right) - g \right\|^2 + \frac{1}{2} \|K(f_n - f_m)\|^2 \\ &\quad + 2\alpha \left\| \frac{f_n + f_m}{2} \right\|^2 + \frac{\alpha}{2} \|f_n - f_m\|^2 \\ &= 2J_\alpha \left(\frac{f_n + f_m}{2} \right) + \frac{1}{2} \|K(f_n - f_m)\|^2 + \frac{\alpha}{2} \|f_n - f_m\|^2. \end{aligned}$$

Since $J_\alpha \left(\frac{f_n + f_m}{2} \right) \geq J_0$ and $\|K(f_n - f_m)\|^2 \geq 0$, it follows that

$$J_\alpha(f_n) + J_\alpha(f_m) \geq 2J_0 + \frac{\alpha}{2} \|f_n - f_m\|^2.$$

Taking the limit as $n, m \rightarrow \infty$, the left-hand side converges to $2J_0$. Consequently, $2J_0 \geq 2J_0 + \frac{\alpha}{2} \lim_{n, m \rightarrow \infty} \|f_n - f_m\|^2$, which implies $\lim_{n, m \rightarrow \infty} \|f_n - f_m\| = 0$ because $\alpha > 0$. Thus, $\{f_n\}$ is a Cauchy sequence in F . Since F is a Hilbert space (and thus complete), there exists an element $f_\alpha \in F$ such that $f_n \rightarrow f_\alpha$.

The functional J_α is continuous (as it is composed of continuous linear operators and norms). Therefore,

$$J_\alpha(f_\alpha) = \lim_{n \rightarrow \infty} J_\alpha(f_n) = J_0.$$

This establishes the existence of a minimizer. The uniqueness follows from the strict convexity of J_α . Specifically, the term $\alpha \|f\|^2$ with $\alpha > 0$ is strictly convex, and $\|Kf - g\|^2$ is convex; the sum of a strictly convex function and a convex function is strictly convex, ensuring at most one minimizer. Alternatively, uniqueness is immediate from the fact that any two minimizers would generate a minimizing sequence with distance zero as shown above.

Secondly, equivalence with the normal equation is demonstrated.

($\Rightarrow$) Suppose f_α is the unique minimizer of J_α . Then for any arbitrary element $f \in F$, we have

$J_\alpha(f) \geq J_\alpha(f_\alpha)$. The difference $J_\alpha(f) - J_\alpha(f_\alpha)$ becomes

$$\begin{aligned}
J_\alpha(f) - J_\alpha(f_\alpha) &= (\|Kf - g\|^2 + \alpha\|f\|^2) - (\|Kf_\alpha - g\|^2 + \alpha\|f_\alpha\|^2) \\
&= (Kf, Kf) - (Kf_\alpha, Kf_\alpha) - 2\operatorname{Re}(Kf, g) + 2\operatorname{Re}(Kf_\alpha, g) + \alpha(f, f) - \alpha(f_\alpha, f_\alpha) \\
&= (K^*Kf, f) - (K^*Kf_\alpha, f_\alpha) - 2\operatorname{Re}(K^*g, f) + 2\operatorname{Re}(K^*g, f_\alpha) + \alpha(f, f) - \alpha(f_\alpha, f_\alpha) \\
&= 2\operatorname{Re}(K^*Kf_\alpha + \alpha f_\alpha - K^*g, f) - 2\operatorname{Re}(K^*Kf_\alpha + \alpha f_\alpha - K^*g, f_\alpha) \\
&\quad - 2\operatorname{Re}(K^*Kf_\alpha, f) + (K^*Kf_\alpha, f_\alpha) + (K^*Kf, f) + \alpha(f_\alpha, f_\alpha) + \alpha(f, f) - 2\operatorname{Re}(f_\alpha, f) \\
&= 2\operatorname{Re}(K^*Kf_\alpha + \alpha f_\alpha - K^*g, f - f_\alpha) + \|K(f - f_\alpha)\|^2 + \alpha\|f - f_\alpha\|^2.
\end{aligned}$$

Since f_α minimizes $J_\alpha(f)$, we substitute $f = f_\alpha + tz$ for any $t > 0$ and $z \in F$, and arrive at

$$2t \operatorname{Re}(K^*Kf_\alpha + \alpha f_\alpha - K^*g, z) + t^2\|Kz\|^2 + \alpha t^2\|z\|^2 \geq 0.$$

Division by $t > 0$ and taking the limit $t \rightarrow 0$ yields

$$\operatorname{Re}(K^*Kf_\alpha + \alpha f_\alpha - K^*g, z) \geq 0,$$

for all $z \in F$. Since z is arbitrary, we can replace z with $-z$, which reverses the sign of the inner product term while keeping the inequality valid. This implies the term must be zero. Thus, we have $K^*Kf_\alpha + \alpha f_\alpha - K^*g = 0$. That is, f_α solves the normal equation.

($\Leftarrow$) Conversely, suppose f_α satisfies the normal equation $(\alpha I + K^*K)f_\alpha = K^*g$. The difference formula reduces to:

$$J_\alpha(f) - J_\alpha(f_\alpha) = \|K(f - f_\alpha)\|^2 + \alpha\|f - f_\alpha\|^2 \geq 0,$$

which confirms that f_α is the global minimizer. $\square$

Theorem 4.4 (Discrepancy Principle). *Let $K : F \rightarrow G$ be linear, compact and one-to-one with dense range in G . Let $Kf = g$ with $f \in F, g \in G, g^\delta \in G$ such that $\|g - g^\delta\| \leq \delta < \|g^\delta\|$. Let the Tikhonov solution f_α^δ satisfy*

$$\|Kf_\alpha^\delta - g^\delta\| = \delta. \tag{B.3}$$

Then $f_\alpha^\delta \rightarrow f$ for $\delta \rightarrow 0$.

Proof. We first prove weak convergence. For any $z \in G$,

$$\begin{aligned}
|(Kf_\alpha^\delta - g, z)| &\leq \|Kf_\alpha^\delta - g\| \|z\| \\
&\leq (\|Kf_\alpha^\delta - g^\delta\| + \|g^\delta - g\|) \|z\| \\
&\leq 2\delta \|z\|.
\end{aligned}$$

This implies that $Kf_\alpha^\delta \rightharpoonup g$, or equivalently $f_\alpha^\delta \rightharpoonup K^{-1}g = f$, as $\delta \rightarrow 0$.

The proof of strong convergence proceeds as follows. The minimization property of the Tikhonov functional implies the inequality $\|Kf_\alpha^\delta - g^\delta\|^2 + \alpha\|f_\alpha^\delta\|^2 \leq \|Kf - g^\delta\|^2 + \alpha\|f\|^2$ for any $f \in F$. From this, the following chain of inequalities is obtained

$$\begin{aligned}\delta^2 + \alpha\|f_\alpha^\delta\|^2 &= \|Kf_\alpha^\delta - g^\delta\|^2 + \alpha\|f_\alpha^\delta\|^2 \\ &\leq \|Kf - g^\delta\|^2 + \alpha\|f\|^2 \\ &= \|g - g^\delta\|^2 + \alpha\|f\|^2 \\ &\leq \delta^2 + \alpha\|f\|^2.\end{aligned}$$

Subtracting δ^2 from both sides and dividing by $\alpha > 0$ yields

$$\|f_\alpha^\delta\| \leq \|f\|.$$

Finally, an analysis of the norm of the difference gives:

$$\begin{aligned}\|f_\alpha^\delta - f\|^2 &= \|f_\alpha^\delta\|^2 - 2\operatorname{Re}(f_\alpha^\delta, f) + \|f\|^2 \\ &\leq 2\|f\|^2 - 2\operatorname{Re}(f_\alpha^\delta, f) \\ &= 2\operatorname{Re}(f - f_\alpha^\delta, f).\end{aligned}$$

Due to the weak convergence established earlier, the right-hand side converges to 0 as $\delta \rightarrow 0$. Thus, $\|f_\alpha^\delta - f\| \rightarrow 0$, which proves the strong convergence $f_\alpha^\delta \rightarrow f$ as $\delta \rightarrow 0$. $\square$

Theorem 4.5 (Convergence of finite element method). *For fixed noise level δ and regularization parameter $\alpha > 0$, we have*

$$\|f_{\alpha,n}^\delta - f_\alpha^\delta\|_F \rightarrow 0, \quad n \rightarrow \infty. \quad (\text{B.4})$$

Proof. We begin by proving weak convergence, and subsequently demonstrate strong convergence.

Weak convergence. Let $P_n : F \rightarrow X_n$ denote the orthogonal projection onto the finite-dimensional subspace $X_n \subset F$, where F is either $L^2(t_{\min}, \Lambda)$ or $H^1(t_{\min}, \Lambda)$. By definition, for any $u \in F$ and all $v \in X_n$, it satisfies $(u - P_n u, v)_F = 0$. Standard finite element approximation results ensure that $\|u - P_n u\|_F \rightarrow 0$ as $n \rightarrow \infty$, for $u \in H^1(t_{\min}, \Lambda)$. Since f_α^δ solves $\alpha f_\alpha^\delta + K^* K f_\alpha^\delta = K^* g^\delta$, and K^* maps G into $C^\infty[t_{\min}, \Lambda]$ due to the smoothing property of the integral kernel, it follows that $f_\alpha^\delta \in H^1(t_{\min}, \Lambda)$. Consequently,

$$\|f_\alpha^\delta - P_n f_\alpha^\delta\|_F \rightarrow 0, \quad n \rightarrow \infty. \quad (\text{B.5})$$

Now f_α^δ is the global minimizer of the functional in F from Eq. (4.16), while $f_{\alpha,n}^\delta$ minimizes the same

functional restricted to X_n as in Eq. (4.21). Hence, $J_\alpha(f_\alpha^\delta) \leq J_\alpha(f_{\alpha,n}^\delta) \leq J_\alpha(P_n f_\alpha^\delta)$. Then,

$$\begin{aligned}
0 &\leq J_\alpha(P_n f_\alpha^\delta) - J_\alpha(f_\alpha^\delta) \\
&= \|K(P_n f_\alpha^\delta - f_\alpha^\delta)\|_G^2 + 2(K(P_n f_\alpha^\delta - f_\alpha^\delta), K f_\alpha^\delta - g^\delta)_G \\
&\quad + \alpha \|P_n f_\alpha^\delta - f_\alpha^\delta\|_F^2 + 2\alpha(P_n f_\alpha^\delta - f_\alpha^\delta, f_\alpha^\delta)_F \\
&\leq \|K\|^2 \|P_n f_\alpha^\delta - f_\alpha^\delta\|_F^2 + 2\|K\| \|P_n f_\alpha^\delta - f_\alpha^\delta\|_F \|K f_\alpha^\delta - g^\delta\|_G \\
&\quad + \alpha \|P_n f_\alpha^\delta - f_\alpha^\delta\|_F^2 + 2\alpha \|P_n f_\alpha^\delta - f_\alpha^\delta\|_F \|f_\alpha^\delta\|_F.
\end{aligned}$$

From Eq. (B.5), we conclude $J_\alpha(P_n f_\alpha^\delta) - J_\alpha(f_\alpha^\delta) \rightarrow 0$, as $n \rightarrow \infty$. Thus, $J_\alpha(f_{\alpha,n}^\delta) \rightarrow J_\alpha(f_\alpha^\delta)$ as $n \rightarrow \infty$, indicating that $f_{\alpha,n}^\delta$ is a minimizing sequence. Because $J_\alpha(f_{\alpha,n}^\delta)$ is convergent, it is bounded. The coercivity $\alpha \|f_{\alpha,n}^\delta\|_F^2 \leq J_\alpha(f_{\alpha,n}^\delta)$ then implies $\|f_{\alpha,n}^\delta\|_F$ is uniformly bounded in n . Consequently, there exists a subsequence f_{α,n_k}^δ and a v in F , such that $f_{\alpha,n_k}^\delta \rightharpoonup v$, as $k \rightarrow \infty$.

Since $K : F \rightarrow G$ is a bounded linear operator, weak convergence in F implies weak convergence in G : $K f_{\alpha,n_k}^\delta \rightharpoonup K v$, as $k \rightarrow \infty$. By the weak lower semicontinuity of the norm, $J_\alpha(v) \leq \liminf_{k \rightarrow \infty} J_\alpha(f_{\alpha,n_k}^\delta)$. Since $J_\alpha(f_{\alpha,n}^\delta) \rightarrow J_\alpha(f_\alpha^\delta)$ as $n \rightarrow \infty$, the same holds for any subsequence. In particular, $\lim_{k \rightarrow \infty} J_\alpha(f_{\alpha,n_k}^\delta) = J_\alpha(f_\alpha^\delta)$. Hence, $J_\alpha(v) \leq \liminf_{k \rightarrow \infty} J_\alpha(f_{\alpha,n_k}^\delta) = J_\alpha(f_\alpha^\delta)$. This shows that v is the minimizer of J_α over F . By Theorem 4.3, the minimizer is unique, so $v = f_\alpha^\delta$. Hence,

$$f_{\alpha,n_k}^\delta \rightharpoonup f_\alpha^\delta, \quad \text{weakly in } F, \quad k \rightarrow \infty. \quad (\text{B.6})$$

Strong convergence. Suppose, for contradiction, that the convergence of norms does not hold, i.e., $\limsup_{k \rightarrow \infty} \|f_{\alpha,n_k}^\delta\|_F > \|f_\alpha^\delta\|_F$. By the weak lower semicontinuity,

$$\begin{aligned}
\|K f_\alpha^\delta - g^\delta\|_G^2 &\leq \liminf_{j \rightarrow \infty} \|K f_{\alpha,n_j}^\delta - g^\delta\|_G^2 \leq \limsup_{j \rightarrow \infty} \|K f_{\alpha,n_j}^\delta - g^\delta\|_G^2 \\
&= \limsup_{j \rightarrow \infty} \left(J(f_{\alpha,n_j}^\delta) - \alpha \|f_{\alpha,n_j}^\delta\|_F^2 \right) < J(f_\alpha^\delta) - \alpha \|f_\alpha^\delta\|_F^2 \\
&= \|K f_\alpha^\delta - g^\delta\|_G^2,
\end{aligned} \quad (\text{B.7})$$

which is a contradiction. Hence, $\limsup_{k \rightarrow \infty} \|f_{\alpha,n_k}^\delta\|_F \leq \|f_\alpha^\delta\|_F$. Together with the weak lower semicontinuity of the norm in F , we have

$$\|f_\alpha^\delta\|_F \leq \liminf_{k \rightarrow \infty} \|f_{\alpha,n_k}^\delta\|_F \leq \limsup_{k \rightarrow \infty} \|f_{\alpha,n_k}^\delta\|_F \leq \|f_\alpha^\delta\|_F. \quad (\text{B.8})$$

Thus, $\|f_{\alpha,n_k}^\delta\|_F \rightarrow \|f_\alpha^\delta\|_F$ as $k \rightarrow \infty$. We obtain strong convergence in F . This completes the proof. $\square$

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